

**Department of Computer Science, College of Science and
Technology
University of Human Development**



Comprehensive study of inter-vlan routing using Router on A Stick and Switched Virtual Interface

**Submitted in partial fulfillment of the requirement of the
BSc degree in Computer Science**

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Abstract

The purpose of this research is to compare the performance of Switched Virtual Interface (SVI) with Router on a Stick (RoAS) using the EVE-NG and GNS3 simulators, as well as the Wireshark tool, for analyzing network metrics such as packet per second, bandwidth utilization, latency, congestion, jitter, packet loss, throughput, download speed, scalability, reliability, cost, and security. The goal of this research is to establish the best alternative for network design and deployment between SVI and RoAS. The study adopts a research technique that comprises configuring a network topology consisting of routers, switches, and virtual computers in both EVE-NG and GNS3 environments. Wireshark was used to run a series of network experiments to evaluate network performance under various traffic circumstances. According to the conclusions of this study, SVI surpasses RoAS in all network metrics except security, where RoAS is marginally superior. In terms of network performance, SVI is determined to be 11 times superior to RoAS. Furthermore, the analysis shows that RoAS is just marginally more secure than SVI. Nonetheless, the benefits of SVI in terms of performance, scalability, dependability, and cost-effectiveness considerably outweigh the small value of RoAS in security. The study finds that SVI is the best alternative for network design and deployment due to its higher performance and cost-effectiveness. Furthermore, with a confidence level of 99 percent, this study exceeds earlier studies in simulation. The study gives unique insights into the comparison of SVI and RoAS in network design and implementation, and the findings can help network engineers, administrators, and researchers make educated network architecture and topology decisions.

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1. Introduction

Inter-VLAN routing, which enables communication between devices on various VLANs, is a crucial component of network architecture. Several methods, including router-on-a-stick, Layer 3 switches with VLAN routing, and Layer 3 switches with Switch Virtual Interfaces (SVIs), can be used to accomplish VLAN routing. Router on a Stick (RoAS) using sub-interfaces can be used to accomplish this. The goal of this article is to present a thorough analysis of inter-VLAN routing utilizing switch virtual interfaces and sub-interfaces. The paper will look at these strategies' advantages and disadvantages, how they were put into practice, and the tools that were utilized to accomplish inter-VLAN routing. The paper will also analyze the various studies done on this subject, outlining the strengths and weaknesses of each [1][2]. An SVI is essentially a virtual interface that is associated with a VLAN on a switch and allows the switch to route traffic between VLANs. SVIs are created to provide a gateway for a VLAN and enable communication between devices in different VLANs. Each SVI is associated with a particular VLAN and has an IP address assigned to it. When a device in one VLAN wants to communicate with a device in another VLAN, the traffic is sent to the SVI of the source VLAN, which then routes the traffic to the destination VLAN's SVI [2][3]. RoAS, which stands for router on a stick, is a networking configuration that allows a single physical router interface to route traffic for multiple virtual local area networks (VLANs). It is a method of inter-VLAN routing that uses a single connection between a router and a switch, typically using a trunk link. In this configuration, the router is connected to a switch that has multiple VLANs configured on it. Instead of using multiple physical router interfaces to connect to each VLAN, a single physical interface on the router is

configured to act as a trunk link. The trunk link carries traffic from all VLANs to the router, where the router performs the routing function and forwards traffic between VLANs[3]. A comprehensive grasp of the underlying technologies and settings is necessary to implement inter-VLAN routing utilizing sub-interfaces and SVIs. To enable effective communication across VLANs, network administrators must be able to establish the virtual interfaces, assign IP addresses, and set up routing protocols. In order to give a thorough analysis of inter-VLAN routing utilizing sub-interfaces and SVIs, this research will look at its advantages, disadvantages, implementation, and related technologies. The paper will also highlight the many studies that have been done on this subject and analyze their strengths and weaknesses [1] [2] [4].

1.1 Aims and objectives of the project.

In the field of network topology, there are several key factors that must be taken into consideration when designing and implementing a network. These include security, reliability, scalability, congestion, performance, bandwidth utilization, delay, and cost.

Security is one of the most critical factors, as networks can be vulnerable to a wide range of threats, including hacking, viruses, and other malicious attacks. It is essential to implement robust security measures to protect the network and its users.

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Performance is also a key consideration, as networks must be able to deliver fast and reliable service to users. Bandwidth utilization and delay are closely linked to performance, as they can impact the speed and responsiveness of the network [5].

Scalability is another crucial factor, particularly in large and complex networks. Networks must be able to grow and adapt to meet changing needs without compromising their overall performance or security [6] [7].

Bandwidth utilization The quantity of network traffic being sent through a network connection at any particular moment is referred to as bandwidth utilization. It is a gauge of a network connection's effectiveness and is stated as a percentage of the total bandwidth available. Network performance can be affected by bandwidth consumption since it can lead to lost packets, delays, and other difficulties [6] [8].

Congestion is a common problem in many networks, particularly those with high levels of traffic. It is essential to manage congestion effectively to ensure that the network can continue to operate efficiently and effectively [9] [10] [11].

Reliability is also important, as network downtime can be costly and disruptive. Networks must be designed to minimize the risk of failure and ensure that they can quickly recover from any outages or other issues [12] [13].

Cost is a critical consideration, particularly for businesses and organizations with limited budgets. It is essential to balance the need for security, reliability, and performance with the cost of implementing and maintaining the network [12] [13].

Security Each device has a unique function inside a network, and different security dangers might apply to different device types. It is crucial to comprehend the security distinctions between a switch layer 3 and a router [14].

2. Literature Review

Some studies that can relate to our work The researchers in this study created a VLAN (virtual local area network) to establish a baseline server and broadband wireless servers, which they then analyzed and examined. They utilized a simulation model that consisted of four buildings and two logical groups of users that used two network applications: FTP and web browsing applications. The OPNET simulator was used in this study[15]. In another study, the researchers created a VLAN (virtual local area network) using the SVI (switched virtual interface) method to evaluate the performance of several protocols, including AODV, OLSR, DSR, TORA, and GRP. They used the OPNET simulator to create FTP and HTTP servers for generating traffic in their simulation model. The study's metrics focused on throughput and delay [16]. This study aimed to evaluate the performance of SVI (switched virtual interface). The researchers used the OPNET simulator to create HTTP, FTP, and database servers for generating traffic in their simulation model. The study's metrics focused on bandwidth, latency, and jitter [17]. Another study was to evaluate the performance of SVI (Switched Virtual Interface) by enhancing its security through the implementation of IPv6. The researchers mainly generated traffic by sending ICMP packets, and the only metric they used in the study was latency [18]. The goal of this research was to build and test a RoAS (Router on a Stick) using the GNS3 simulator. The researchers used a command to transmit 25 packets to generate traffic. The assessment criteria for the study were confined to latency and packet loss. Latency is the time it takes between sending and receiving packets, whereas packet loss is the number of packets that fail to reach their destination. The purpose of the study was to evaluate the RoAS's capacity to manage network traffic by assessing its latency and packet loss rates. The GNS3 simulator was used to build a virtual network environment, allowing the researchers to test the RoAS's performance in a controlled environment[19]. The research aimed to create a RoAS model using the Packet Tracer simulator.

In order to accomplish this, network traffic was simulated through the use of ping commands. The focus of the study was limited to measuring latency and jitter metrics. By utilizing these metrics [20]. The primary objective of this research study was to develop VLANs using the RoAS approach in the Packet Tracer software. To evaluate the performance of the network, network traffic was simulated using the ping command. The study solely focused on measuring the latency and jitter metrics to assess the quality of service provided to network users [21]. Another research study was conducted with the specific goal of creating virtual LANs (VLANs) using the switched virtual interface (SVI) approach within the GNS3 simulator. The network traffic was generated through the utilization of the ping command, and the study focused solely on measuring the latency and jitter metrics [22]. Another research study was undertaken with the specific goal of utilizing the EtherChannel LACP-type technology as a design approach for achieving link redundancy and load balancing. The study was conducted within the GNS3 simulator using the SVI approach. The network traffic was generated through the utilization of the ping command on a significant number of PCs. The study solely focused on measuring latency and packet loss metrics [23].

2.1 Pros and cons

Pros:

- The studies utilize simulation models to test and evaluate network performance in a controlled environment, allowing for accurate measurements of network metrics.
- The studies focus on different network technologies and protocols, providing a comprehensive understanding of their performance and capabilities.
- The studies use various simulators such as OPNET, GNS3, and Packet Tracer, demonstrating the flexibility and versatility of simulation software in testing network performance.
- The studies provide valuable insights and recommendations for network designers and administrators to improve network performance, such as enhancing security and implementing link redundancy.

Cons:

- The studies are limited to simulation environments, and the results may not fully reflect the actual network performance in real-world scenarios.
- Some studies focus on specific metrics, limiting the scope of their evaluations.
- The studies may not consider the impact of various factors, such as network size and topology, on network performance.

3. Methodology

Planning, obtaining requirements, creating, implementing, testing, and sustaining were all stages of the research process. During the planning stage, the objectives, research subjects, and scope of the study were decided. In addition, the standards for selecting SVI configurations and sub-interfaces were established. During the requirements gathering phase, the hardware and software necessary to design network topologies with sub-interfaces and SVI configurations were determined [1] [2]. Wireshark was used to define the performance metrics as well as the monitoring techniques for throughput, latency, packet loss, and so on. Network topologies were generated throughout the design phase using the simulation tools EVE-NG and GNS3. SVI and sub-interfaces were set on routers and switches based on the standards specified during the planning stage. Sub-interfaces and SVI parameters were then added to the proposed network topologies during the implementation stage. Wireshark, a network monitoring tool, was installed to collect performance data. The performance metrics created during the requirements collection phase were used to measure the performance of the sub-interfaces and SVI configurations throughout the testing phase. To determine the importance of any differences observed between the sub-interfaces and SVI configurations, statistical analysis techniques were applied to the gathered performance data. Throughout the maintenance period, the data was meticulously documented, and recommendations for more research or network design improvements were made. The waterfall method ensured that each phase of the research process was completed correctly before moving on to the next. It used a structured strategy for the study process and provided a methodical and rigorous approach [24] [25] [26].



Figure 1 : Gns3, Eve-ng, Wireshark

3.1 Requirement Analysis

Comprehensive set of criteria summarized in three tables: Table 1, Table 2, and Table 3. These tables serve as useful references, encompassing the aspects required for the effective execution of our project. Each table provides detailed insights into many parts of the project, ensuring that all critical factors are handled.

Tool Requirement	Tools and Apps
Network Simulator	GNS3 and EVE-NG
Gathering Information	CCNA , CCNP , Reaserchpaper , ...etc
Installing Virtual Images	Router, Switch layer 2 , switch layer 3, Windows 7 , windows server, kali linux
Monitoring	Wireshark , cisco command
Graph Tool	Canva , wireshark

Table 1: Tool and apps

Vlan Requirement	Configuration
Configuring	SVI and Router On a Stick
Accessing Vlans to PCs	Switch port mode access Vlans
Trunking Switches	Switch mode Trunk
Packet send	Ping to other Vlans
Create Traffic (Load)	Configuring FTP server , Sending Attacks

Table 2: Vlan requirement

Network Metric Requirement	Detail
Latency	Amount of time it takes for a data packet to travel from its source to its destination
Jitter	variation in the delay of packet arrival in a network connection
Packet loss	Failure of a data packet to reach its intended destination
Bandwidth utilization	Amount of data that is being transferred over a network connection
Download speed	The rate at which data is transferred from a remote server to a user's device.

Congestion	how many packets can be sent at one time
Throughput	Amount of data that is successfully transmitted over a network connection within a given period of time.
Scalability	the ability of a network to scope with increasing workloads
Reliability	Deliver data packets without errors or interruptions.
Cost	Which one cheaper SVI or Router On a Stick
Security	Router and switch L3 each has unique security

Table 3: Network analysis Requirement

3.2 Key Technologies

Network technology includes routers, switches, cables, wireless access points, and hubs. It also includes protocols such as Ethernet and TCP/IP that are used to set up connections between different types of devices on the network. Network technology is a broad term that includes a variety of hardware and software components that allow devices to connect with one another through a network. Network technology components are widely categorized into two types: physical and logical. Physical components include routers, switches, cables, wireless access points, and hubs, which provide physical connectivity between network devices. These devices enable devices to communicate by sending data in the form of electrical or optical signals. Routers are network devices that allow networks to communicate by passing data packets between them. They are in charge of routing data traffic between devices on various networks and can prioritize traffic based on its relevance. Switches, on the other hand, are network devices that link devices. They are in charge of guiding data traffic between network devices and can also prioritize traffic based on its relevance. Cables are used to physically link network equipment. Ethernet cables, fiber-optic cables, and coaxial cables are examples of several types of cables, each with its own set of properties and applications [1][2]. Network technology, in addition to physical components, incorporates intellectual components such as protocols. Protocols are collections of rules and processes that regulate how data is sent over a network. Ethernet and TCP/IP are the two most widely used protocols in network technology. Ethernet is a protocol that governs data transmission across a wired network. It defines how data is encoded and transported, as well as how devices identify and recover from transmission faults. TCP/IP, on the other hand, is a collection of protocols that govern data transmission across the internet. It is in charge of routing data between devices on various networks and includes capabilities such as error detection and repair, data fragmentation, and reassembly [27].

3.4 Usability

The implementation of a project integrating VLAN Switched Virtual Interfaces (SVI) and Router on a Stick (RoAS) is primarily intended for professional network users with expertise in network architecture, configuration, and administration. These users possess a comprehensive understanding of network protocols and are proficient in working with network hardware and software. This type of project may be particularly valuable in large or complex networks where traffic segmentation is critical for enhancing performance, security, or other reasons. VLAN SVIs can be used to isolate traffic within a specific VLAN, preventing it from interfering with other VLAN traffic. Additionally, professional network users may benefit from the flexibility and scalability of VLAN SVIs and RoAS. By creating separate VLANs, traffic can be divided into multiple groups based on department, function, or security level. This can help mitigate the consequences of security breaches, reduce network congestion, and improve overall network performance. Ultimately, the applicability of a project incorporating VLAN SVIs and RoAS is determined by the network's specific needs and requirements, as well as the network administrator's abilities and expertise. Professional network users with experience in network design and maintenance may find this type of project useful in improving network speed, scalability, and security [3][4][28].

4. Design and Implementation

4.1 Design

EVE-NG and GNS3 both use this design. As shown in Figure 2, Figure 3, Figure 4, and Figure 5, The network topology includes SVI and RoAS VLANs, with a router and switch layer 3 at the top, allowing for inter-VLAN routing. The topology design includes an FTP server on the right side that is linked to both the network and the Internet. The computers used in this investigation are running the Windows 7 operating system. It is worth noting that all of the machines in this topology are completely working and have their own pictures, guaranteeing that the study results are correct and dependable. The need for adopting a suitable topology design to support smooth network operations and good inter-VLAN routing is emphasized in the study paper. The addition of a working FTP server and suitably configured PCs confirms that the established topology design is successful [24][25][26].

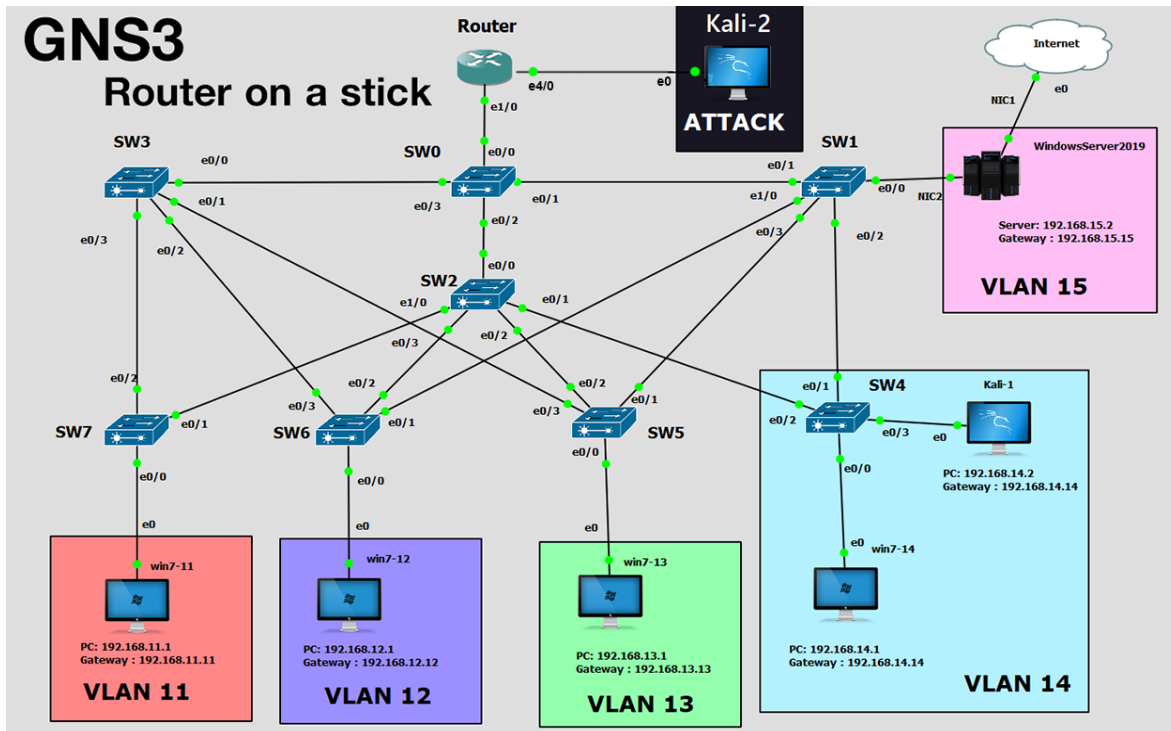


Figure 2 : Router on a stick Design in GNS3

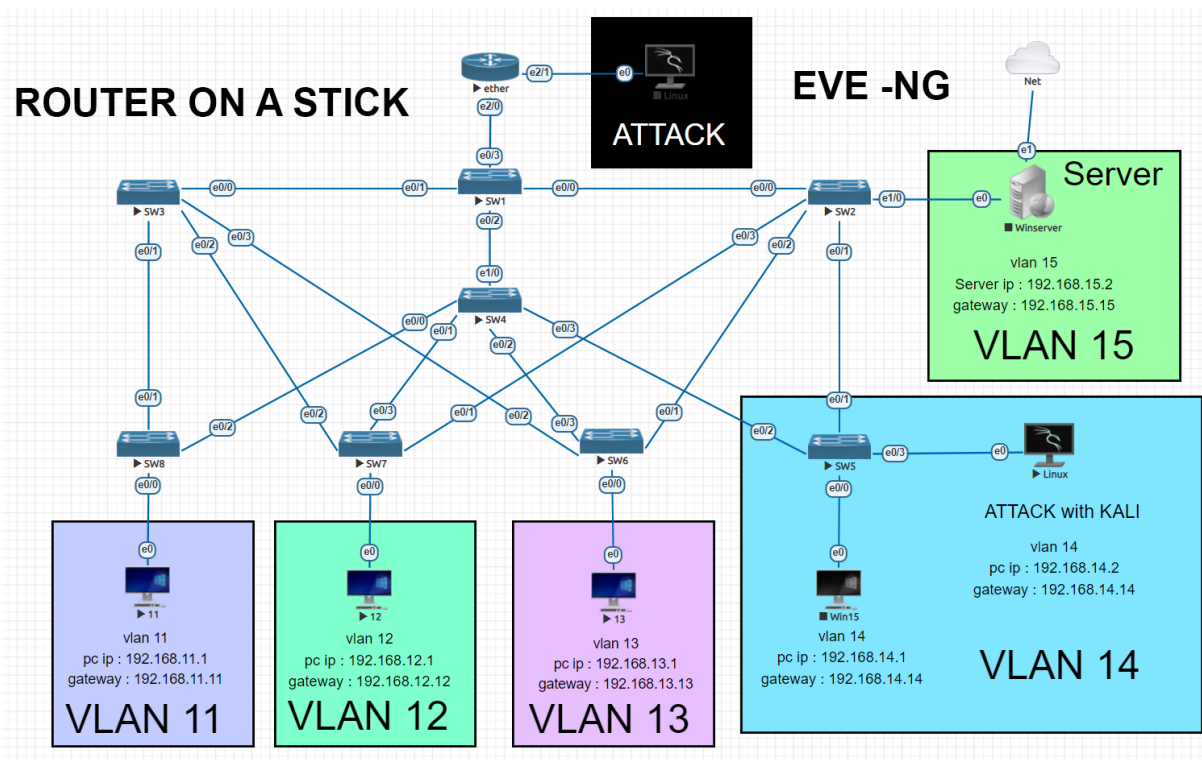


Figure 3 : Router on a stick Design in EVE-NG

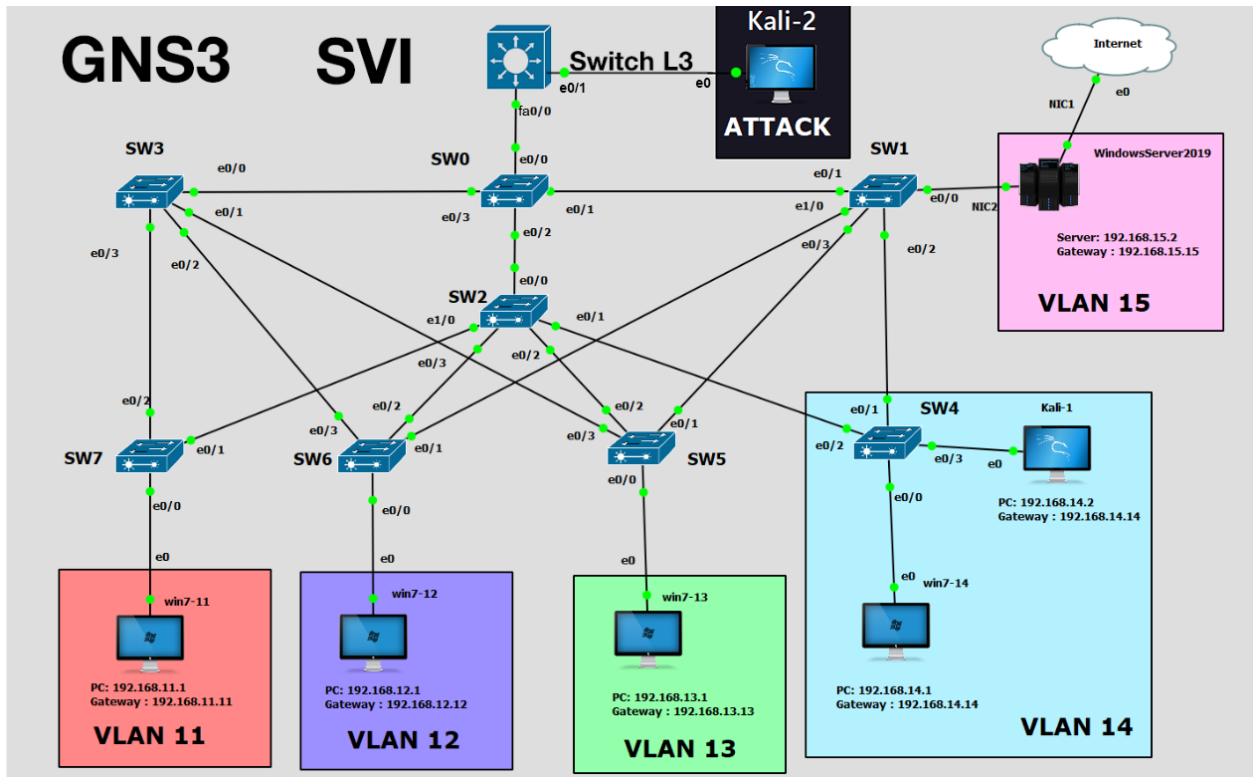


Figure 4 : Switched Virtual Interface Design in GNS3

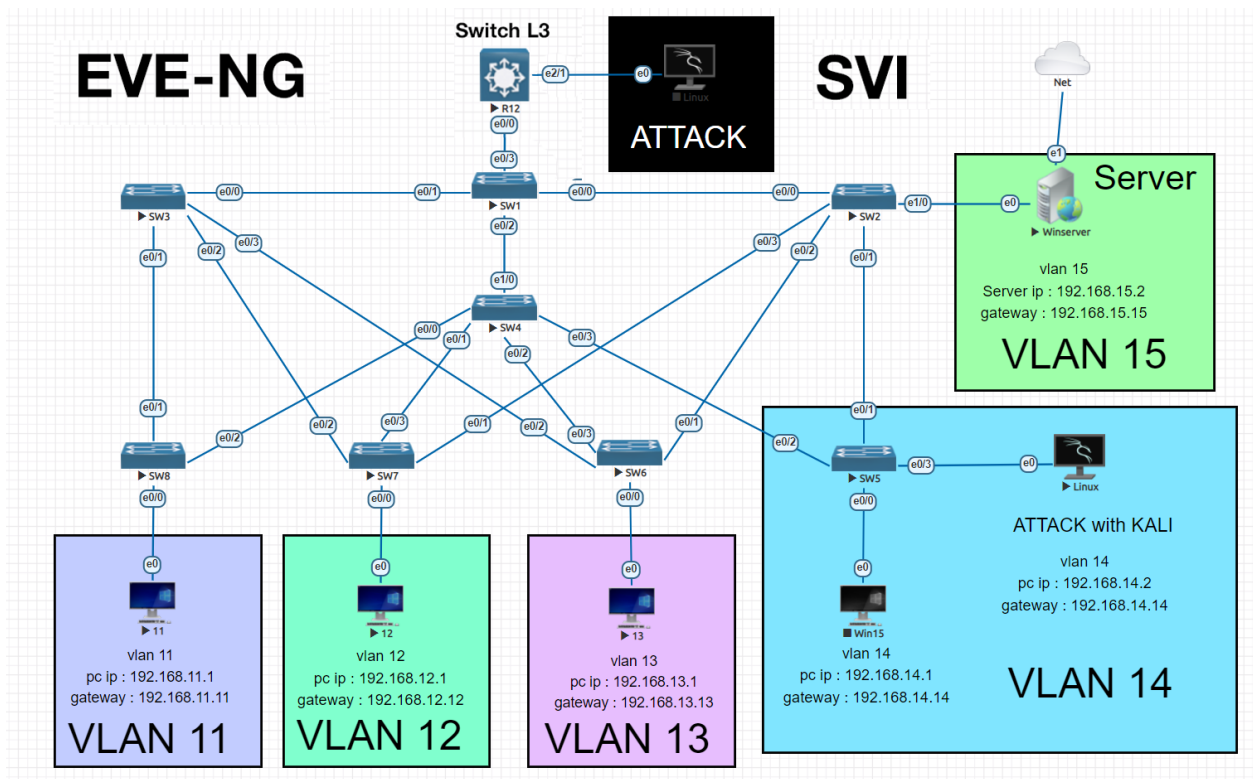


Figure 5 : Switched Virtual Interface Design in EVE-NG

4.2 Implementation

Tables 4 and 5 give a detailed picture of the devices, their corresponding ports, and assigned responsibilities based on the information shown in Figures 2 and 4 of the GNS3 simulation. Table 4 shows a list of devices, together with their related ports and functionality. Table 5 also includes a detailed listing of all IP addresses used in the network setup.

Device	Port	Task
Router	e1/0 , e4/0	Ip routing
Switch Layer3	fa0/0 , e0/1	Ip routing
SW0	Range e0/0-3	Trunking
SW1	Range e1/0-3 , e1/0 trunk e0/0 Vlan 15	Trunking and Switch port access Vlan 15
SW2	Range e0/0-3 , e1/0	Trunking
SW3	Range e0/0-3	Trunking
SW4	e0/0 , e0/3	Switch port access Vlan 14
SW5	e0/0	Switch port access Vlan 13
SW6	e0/0	Switch port access Vlan 12
SW7	e0/0	Switch port access Vlan 11
WindowsServer2019	NIC2	End device , Vlan 15
win7-14	e0	End device , Vlan 14
win7-13	e0	End device , Vlan 13
win7-12	e0	End device , Vlan 12
win7-11	e0	End device , Vlan 11
Kali-1	e0/3	END device ,vlan 14 , Attacker
kali-2	e0	Attacker

Table 4: List of Device ports and their task

Device or Port	IP Address	Gateway	Subnet mask
Router e1/0.11	192.168.11.11		255.255.255.0
Router e1/0.12	192.168.12.12		255.255.255.0
Router e1/0.13	192.168.13.13		255.255.255.0
Router e1/0.14	192.168.14.14		255.255.255.0
Router e1/0.15	192.168.15.15		255.255.255.0
Router e4/0	192.168.2.1		255.255.255.0
Switch L3 Vlan 11	192.168.11.11		255.255.255.0
Switch L3 Vlan 12	192.168.12.12		255.255.255.0
Switch L3 Vlan 13	192.168.13.13		255.255.255.0
Switch L3 Vlan 14	192.168.14.14		255.255.255.0
Switch L3 Vlan 15	192.168.15.15		255.255.255.0
WindowsServer2019	192.168.15.2	192.168.15.15	255.255.255.0
win7-14	192.168.14.1	192.168.14.14	255.255.255.0
win7-13	192.168.13.1	192.168.13.13	255.255.255.0
win7-12	192.168.12.1	192.168.12.12	255.255.255.0
win7-11	192.168.11.1	192.168.11.11	255.255.255.0
kali-1	192.168.14.2	192.168.14.14	255.255.255.0
kali-2	192.168.2.10	192.168.2.1	255.255.255.0

Table 5: List of IP address

4.2.1 Topology Configuration

4.2.1.1 Configuring Vlans

The configuration is several parts you can find the full detail in the(Appendix)

1-Configuring Router on a stick and Switched virtual interface.

2-Creating Vlans on all switches.and Trunking .

3-Accessing ports to the Vlans.

4.2.1.2 Configuring FTP server

When it comes to producing network traffic, setting up an FTP server is a viable and practical choice. A great deal of data is transported across the network when a large video file is downloaded to the server. This data transmission creates traffic, which may be used to assess the network's capacity and performance. Furthermore, users may replicate real-world network usage conditions by accessing the video file from additional VLAN PCs, allowing for a more realistic performance assessment. As a result, putting up an FTP server and transmitting a huge video clip is a great way to generate network traffic and evaluate the network's capabilities [15] [16]. Figures 6 and 7 demonstrate the procedure of configuring a Windows FTP server. Following that, in Figures 8 and 9, the verification process is carried out to confirm the FTP server's appropriate operation.

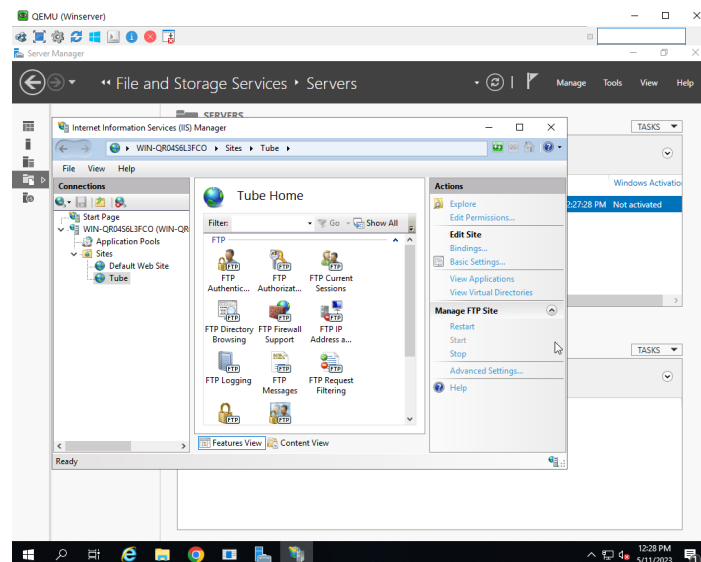


Figure 6 : FTP server

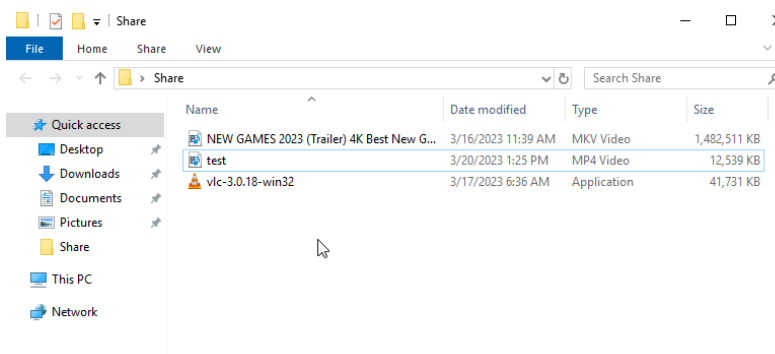


Figure 7 : Files in the Shared folder (FTP server)

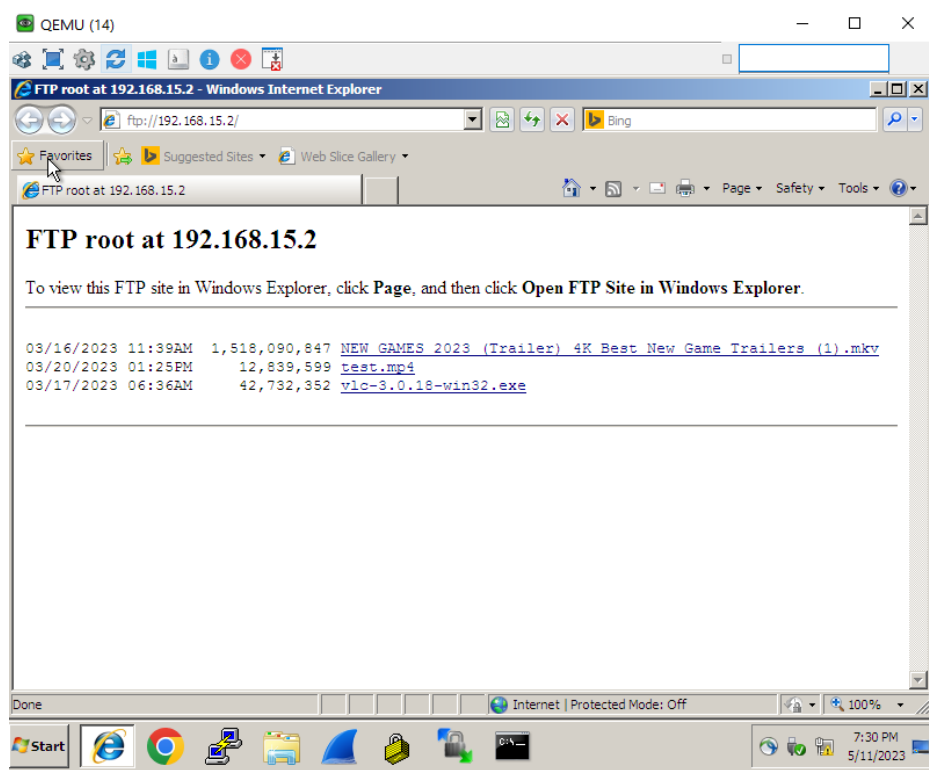


Figure 8 : Checking FTP server in other Vlans PC

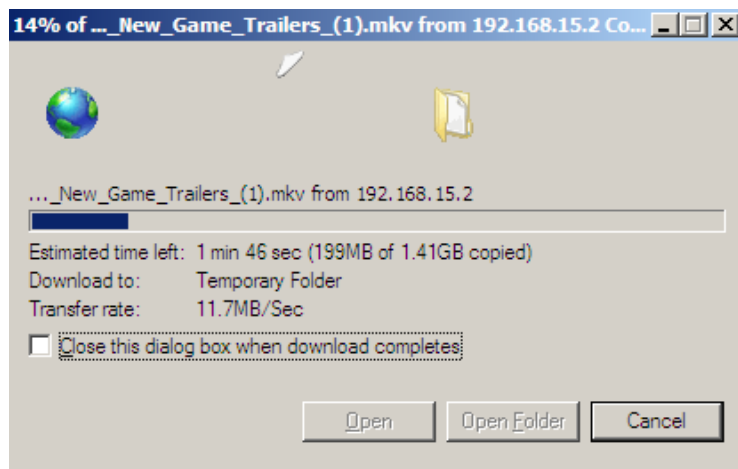


Figure 9 : Downloading Files in other Vlans PC

4.2.2 Performance

When creating and establishing VLANs in a network, the two most frequent techniques for allowing VLAN communication are via a router on a stick and via a Switch Virtual Interface (SVI). It is necessary to examine the performance of both strategies in order to identify which may be optimal for a specific network environment. Many tests may be performed to evaluate the performance of a router on a stick vs. an SVI using various tools such as network device "display" commands and Wireshark packet capture software. Network administrators can use the "display" commands to see the status and configuration of network devices and interfaces, while Wireshark can record and analyze network traffic. Every capture is 1 minute long. To begin, the "display interfaces" command on both the router on a stick and SVI may be used to check the interface status, including any problems or packet drops. This can help identify any possible bottlenecks or interface issues that are harming performance. Following that, the "show ip route" command on the router on a stick may be used to examine the routing table, while the "show vlan" command on the switch with the SVI can be used to view the VLAN setup. This can aid in identifying any inconsistencies or misconfigurations that may be affecting performance. We created some charts manually for better understanding. Lastly, to investigate network traffic between VLANs, a Wireshark capture may be performed on both the router on a stick and the SVI. This can assist in identifying any latency or delay issues as well as potential network congestion causes. Network managers may make an educated judgment about whether the option of a router on a stick or SVI is ideal for their network environment by comparing the results of these tests. Finally, the option will be determined by the network's unique requirements and resources, and a careful review of both techniques will assure maximum performance and efficiency [5] [25] [29] [30].

4.2.2.1 Packet per second

In both GNS3 and EVE-NG network simulation environments, we used the Wireshark tool to calculate the maximum number of packets that may be delivered in Router on a Stick (ROAS) and Switched Virtual Interface (SVI) setups. We used Wireshark to record and analyze network data in order to determine the maximum packet limit for each setup. This information is crucial for network managers to understand their networks' performance capabilities and guarantee optimal utilization. Wireshark results were critical in providing a full and trustworthy examination of network performance in our investigation. In figures 10 and 11, they represent RoAS, and in figures 12 and 13, they represent switched virtual interfaces, it appears that the Switched Virtual Interface (SVI) approach has a higher packet sending rate compared to the Router on a Stick approach. Additionally, the EVE-NG simulator appears to have a higher packet-sending rate than the GNS3 simulator [25].

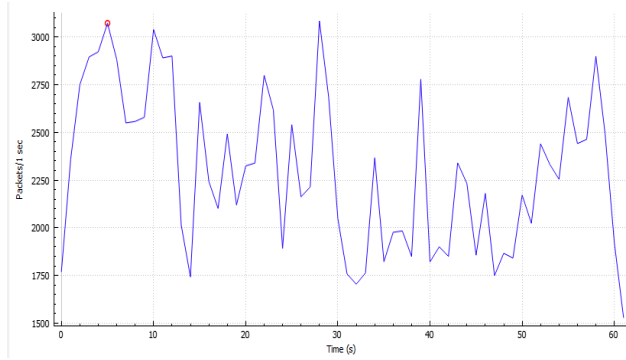


Figure 10: Packet per second RoAS (EVE-NG)

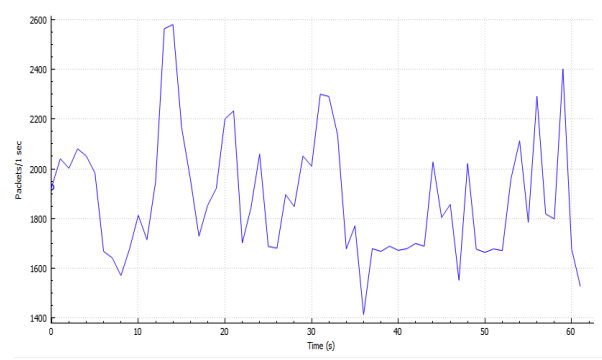


Figure 11: Packet per second RoAS (GNS3)

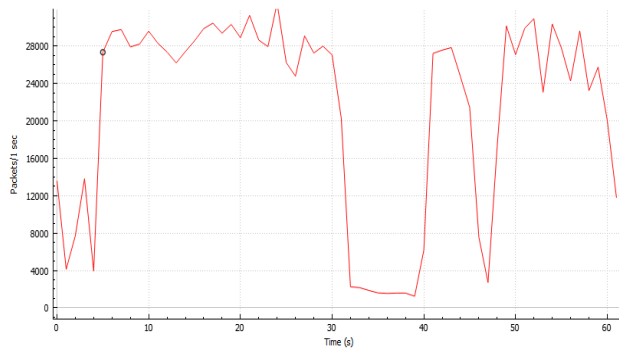


Figure 12 : Packet per second SVI (EVE-NG)

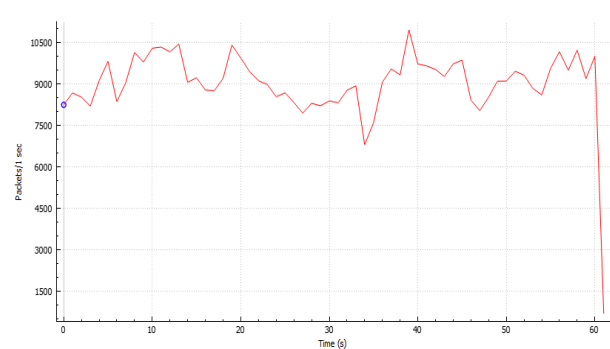


Figure 13 : Packet per second SVI (GNS3)

4.2.2.2 Latency and jitter

Jitter is a variation in the delay of packet arrival in a network connection. The ping command is used in network measurement to measure latency, jitter, and packet loss. Figures 14 and 15 demonstrate this. On the virtual PCs, this is accomplished by sending 30 packets to a destination and assessing the results. To acquire a credible measurement of the network's behavior, the operation is done several times. The output of the ping command is shown below, which provides information about the round-trip time, the variation in the round-trip time (jitter), and the number of packets lost during transmission [19] [20] [21] [41]. in the figures 16, 17, 18, and 19 it shows the jitters on each method.

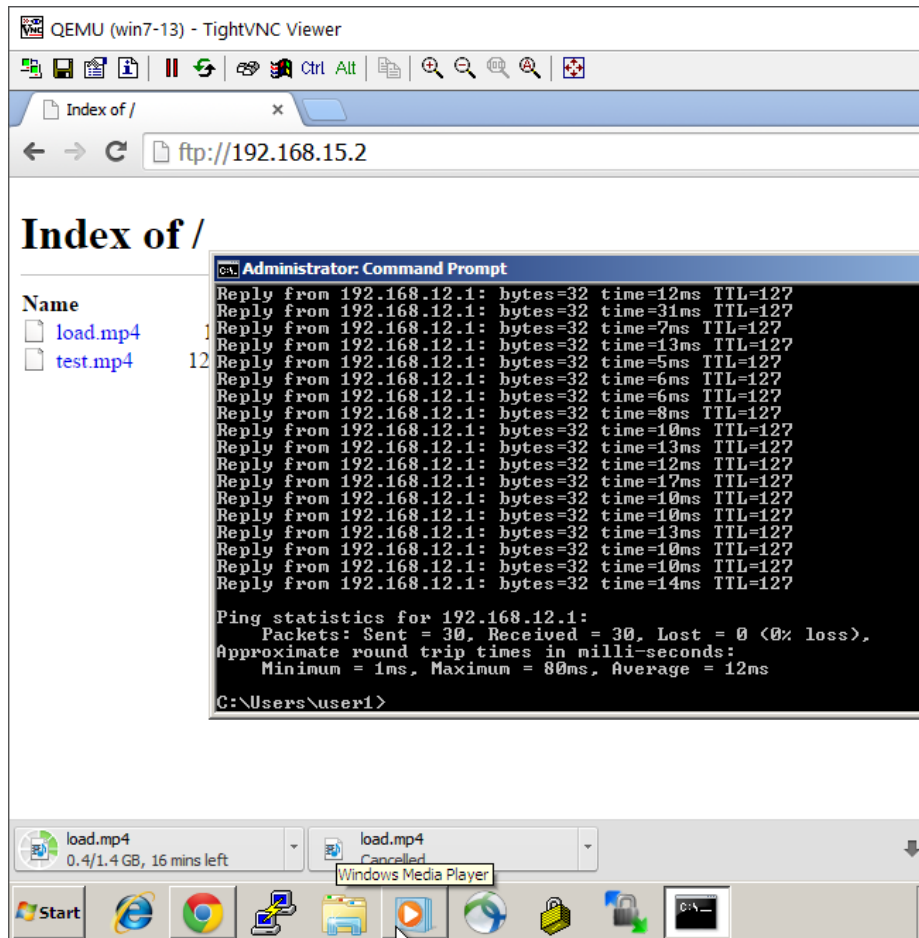


Figure 14 : Attempt 1 (Jitter, Packet loss , Latency)

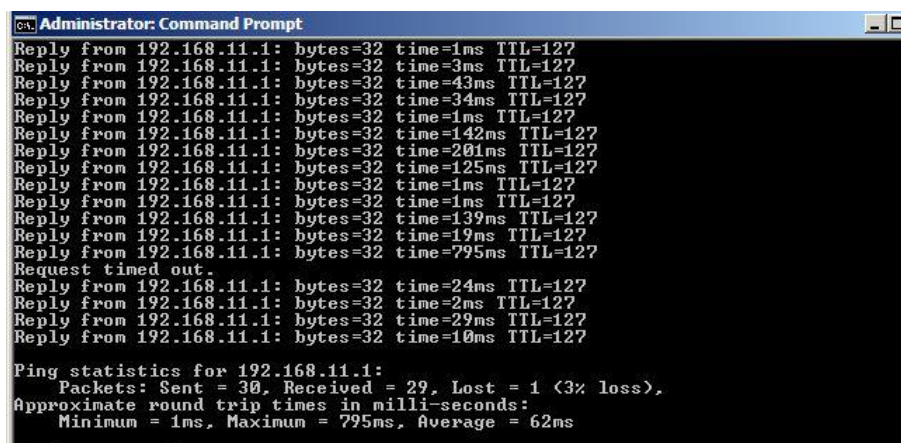
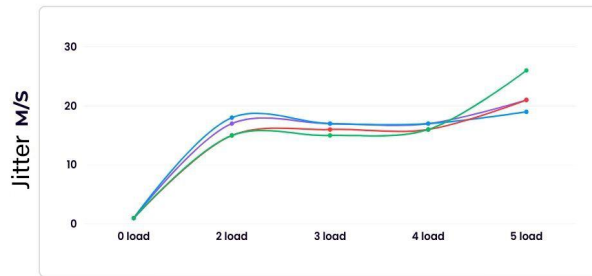


Figure 15 : Attempt 2 (Jitter, Packet loss , Latency)

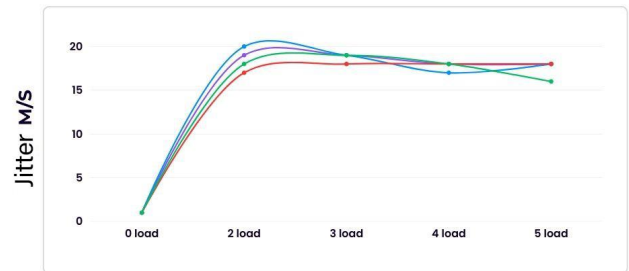
Jitter ON Router EVE-ng



PC 1 PC 2 PC 3 PC 4

Figure 16 : Jitter Router on a stick (EVE-NG)

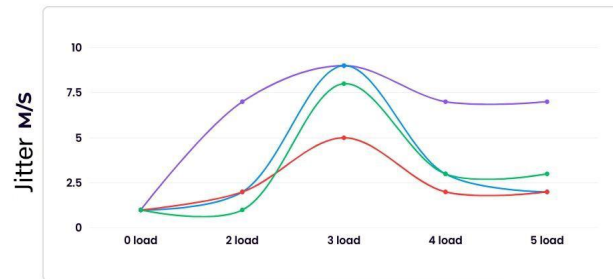
Jitter ON Router GNS3



PC 1 PC 2 PC 3 PC 4

Figure 17 : Jitter Router on a stick (GNS3)

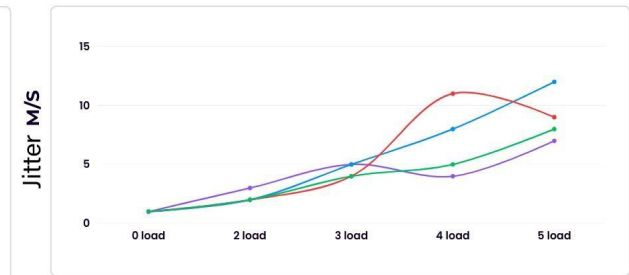
Jitter ON SVI EVE-ng



PC 1 PC 2 PC 3 PC 4

Figure 18: Jitter SVI (EVE-NG)

Jitter ON SVI GNS3



PC 1 PC 2 PC 3 PC 4

Figure 19 :Jitter SVI (GNS3)

The graphical representation of figures 16, 17, 18, and 19. compared the performance of two methods for linking several VLANs on a network device: switched virtual interface (SVI) and router on a stick (ROAS). Based on our findings, SVI outperforms the router on a stick in terms of latency and jitter. Our research found that SVI, which permits inter-VLAN communication to take place within the switch itself, can minimize overall network latency and jitter when compared to a router on a stick. This is due to the fact that a router on a stick requires packets to be routed through a separate physical router for inter-VLAN connectivity, which adds additional latency and jitter. Overall, SVI may be a better alternative than ROAS when network performance is crucial, particularly when there are a significant number of VLANs or substantial network traffic. But, like with any network design decision, it is critical to analyze all considerations and select the technique that best matches the network's unique requirements [15] [16] [17] [41].

4.2.2.3 Throughput

the quantity of data that can be sent via a network in a given amount of time [42]. It is a measure of the network's ability to efficiently transport data. Typically, throughput is stated in bits per second. After monitoring, the results are shown in Figures 20, 21, 21, 22, and 23.

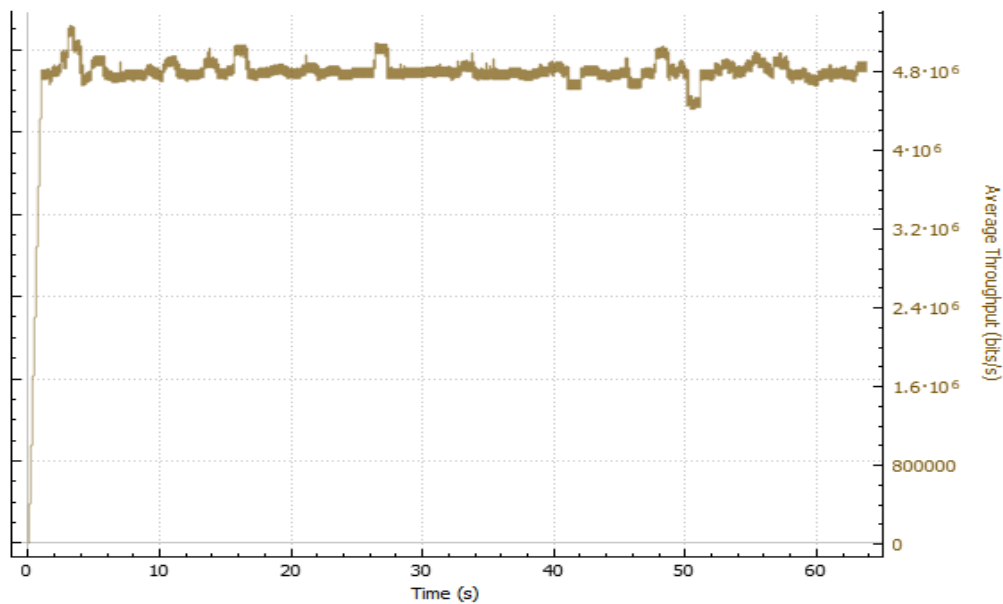


Figure 20: Throughput Router On a Stick (EVE-NG)

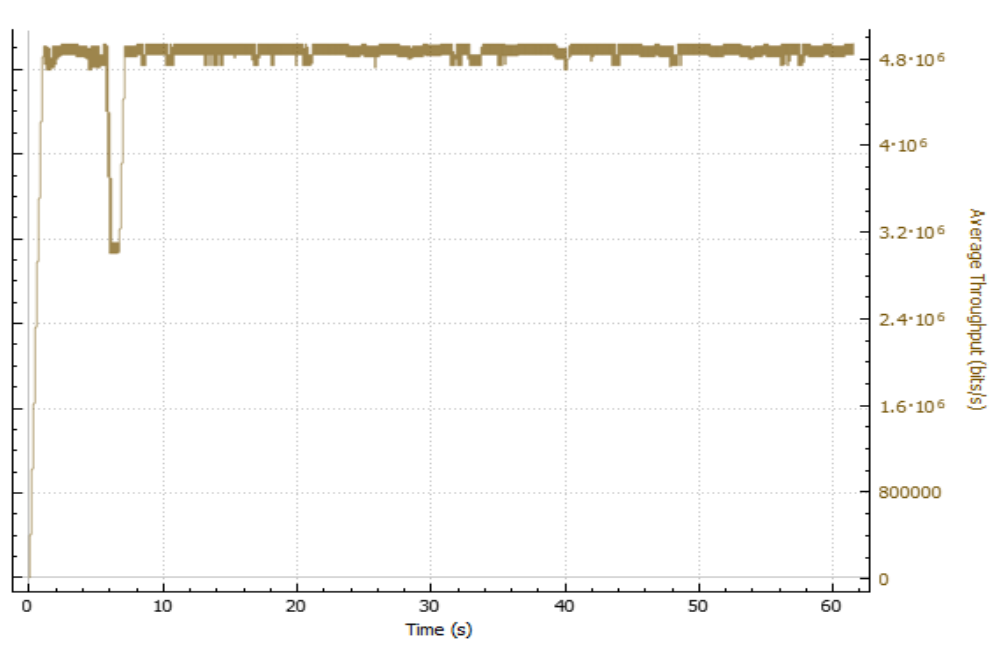


Figure 21: Throughput Router On a Stick (GNS3)

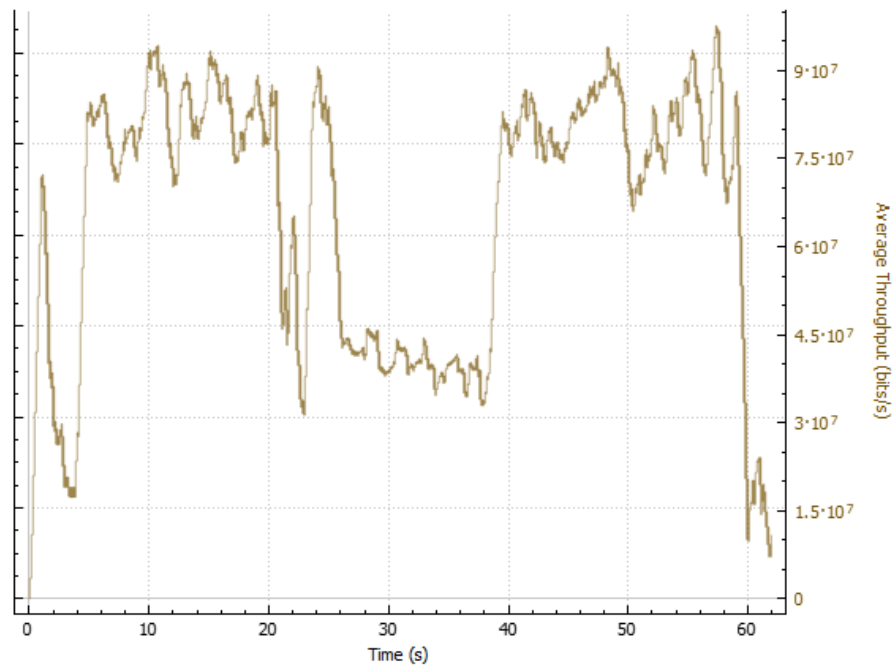


Figure 22: Throughput SVI (EVE-NG)

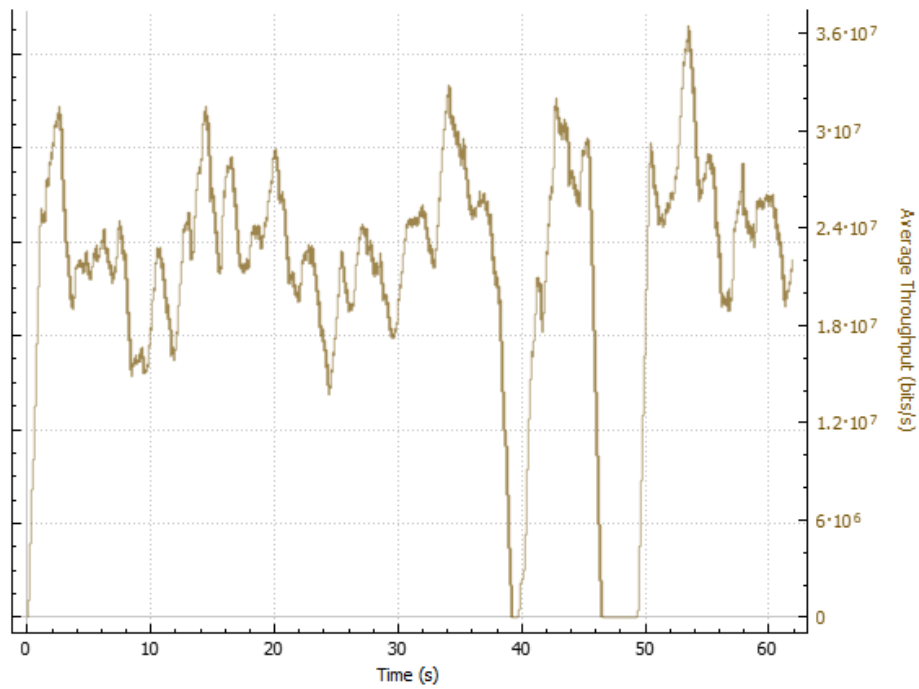


Figure 23: Throughput SVI (GNS3)

According to Figures 20, 21, 22 and 23, there are two methods for linking several VLANs on a network device, switched virtual interface (SVI) and router on a stick (ROAS). According to our findings, SVI outperforms Router on a Stick in terms of throughput. SVI, which permits inter-VLAN communication to take place within the switch itself, can deliver greater throughput than a router on a stick. This is due to SVI's ability to leverage the switch's hardware-based forwarding, which is geared for managing high amounts of traffic. ROAS, on the other hand, is dependent on the router's software-based forwarding, which might be slower and less efficient [42].

4.2.2.4 Download Speed

To conduct a realistic network download speed test, download data from an FTP (File Transfer Protocol) server. Because users generally download a range of files of varying sizes and types, this strategy provides for a more accurate simulation of real-world network consumption patterns. The test is less vulnerable to restrictions such as bandwidth limiting or server-side caching because files are downloaded over the FTP server. Furthermore, FTP servers are particularly intended for file transfers, guaranteeing that the test results are trustworthy and constant. As a result, downloading data from an FTP server is a suggested method for performing realistic and accurate network download speed testing [15] [16] [17] [18]. According to Figures 24 and 25, the Switched Virtual Interface (SVI) approach is significantly faster when it comes to downloading. However, in terms of speed, EVE-NG outperforms GNS3.

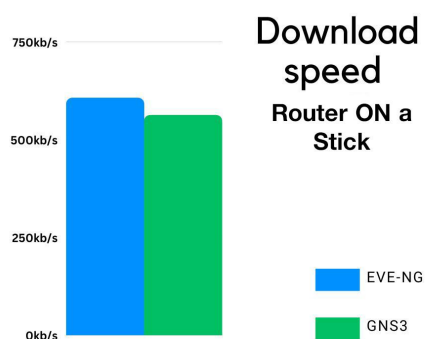


Figure 24 : Download speed Router On a Stick

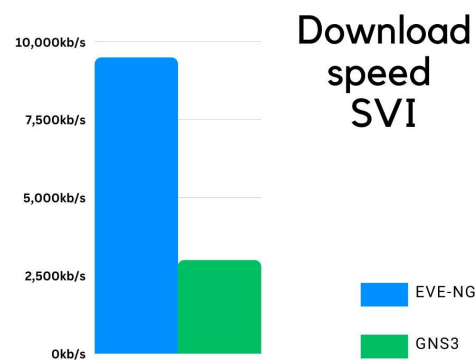


Figure 25: Download speed SVI

4.2.3 Bandwidth utilization

The greatest data transfer rate that a network connection or link can sustain is measured as bandwidth. Bandwidth is measured in megabits per second (Mbps). It specifies the potential maximum amount of data that may be transferred through a network [25] [41]. According to Figures 26 and 27, SVI is much better for Bandwidth utilization but on EVE-NG is much faster than GNS3.

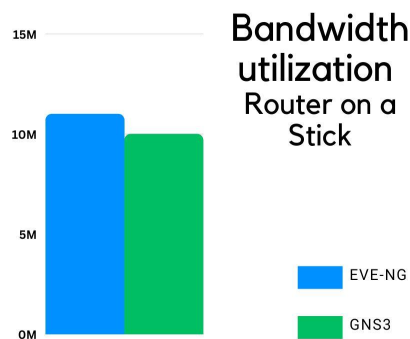


Figure 26: Bandwidth utilization Router on a stick

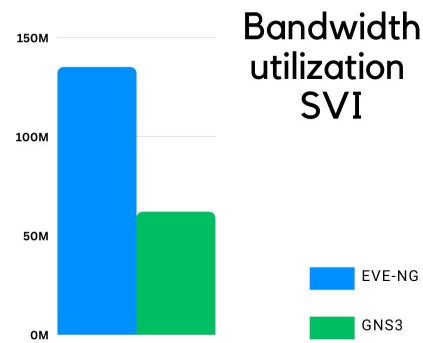


Figure 27: Bandwidth utilization SVI

4.2.4 Congestion

Congestion in a network occurs when the demand for network resources exceeds the available capacity, resulting in a reduction in performance or, in extreme cases, a full breakdown of communication. It arises when there is an imbalance between the quantity of data being transferred and the network's ability to efficiently process that data [30]. According to Figures 28, 29, 30, and 31, Switched Virtual Interface (SVI) surpasses Router on a Stick in terms of network congestion management (ROAS). According to our findings, SVI, which enables inter-VLAN communication within the switch, provides improved congestion management capabilities by spreading traffic more efficiently over numerous VLANs, reducing the load on any one VLAN. ROAS, on the other hand, relies on a single physical interface to manage all inter-VLAN traffic, which increases the danger of network congestion and bottlenecks, especially when dealing with huge amounts of traffic or a large number of VLANs. As a result of our findings, we propose that SVI be considered the best choice for effective congestion control in network architecture. And EVE-NG is better [30] [31] [32].

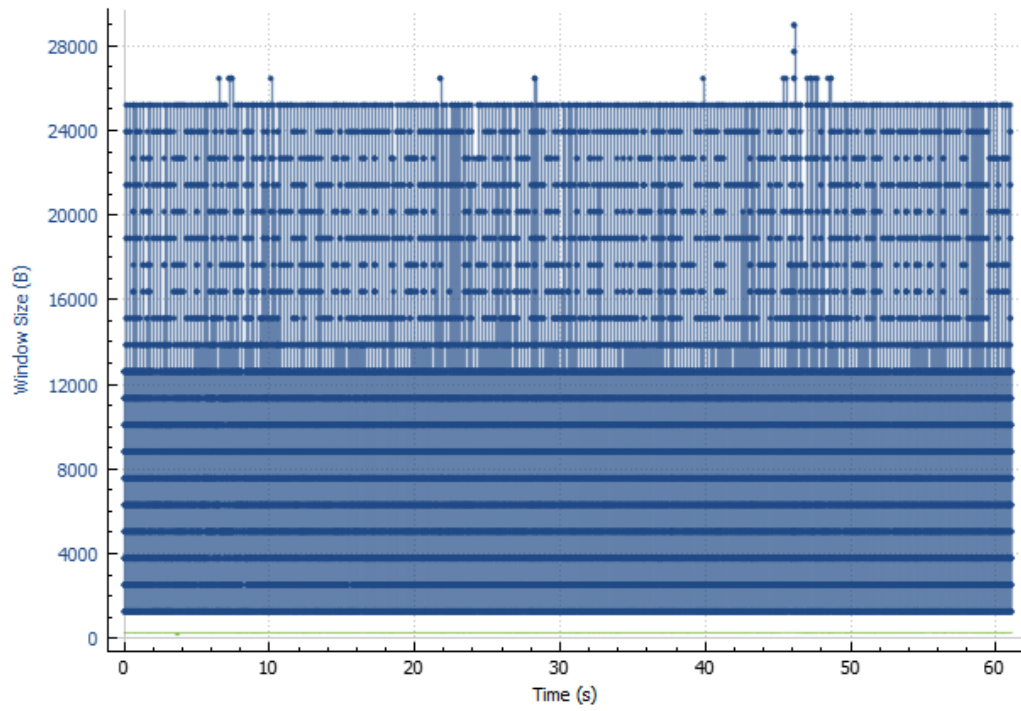


Figure 28 : Congestion Router On a Stick (EVE-NG)

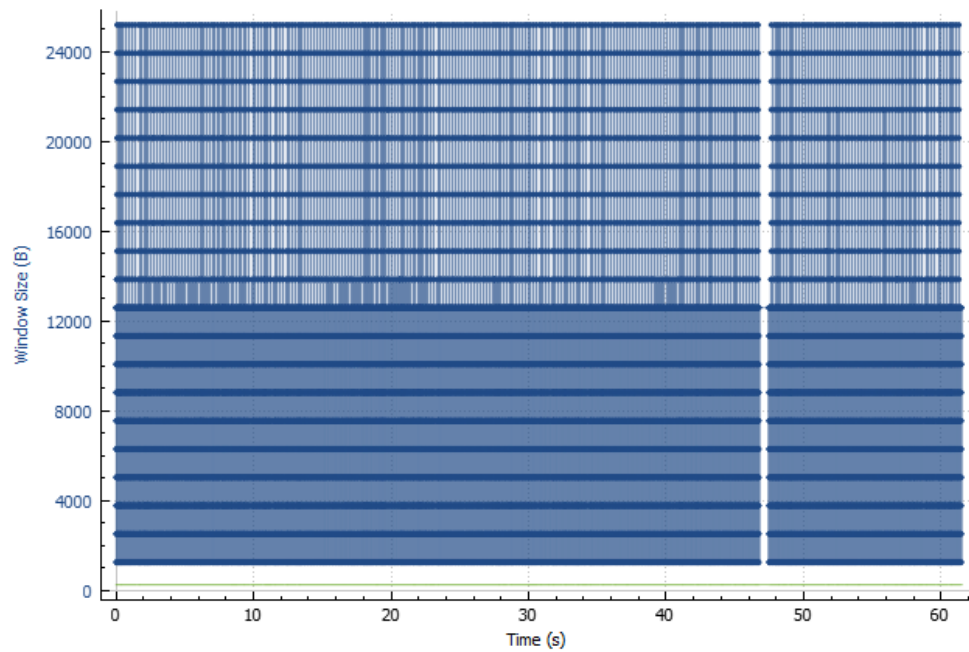


Figure 29: Congestion Router On a Stick (GNS3)

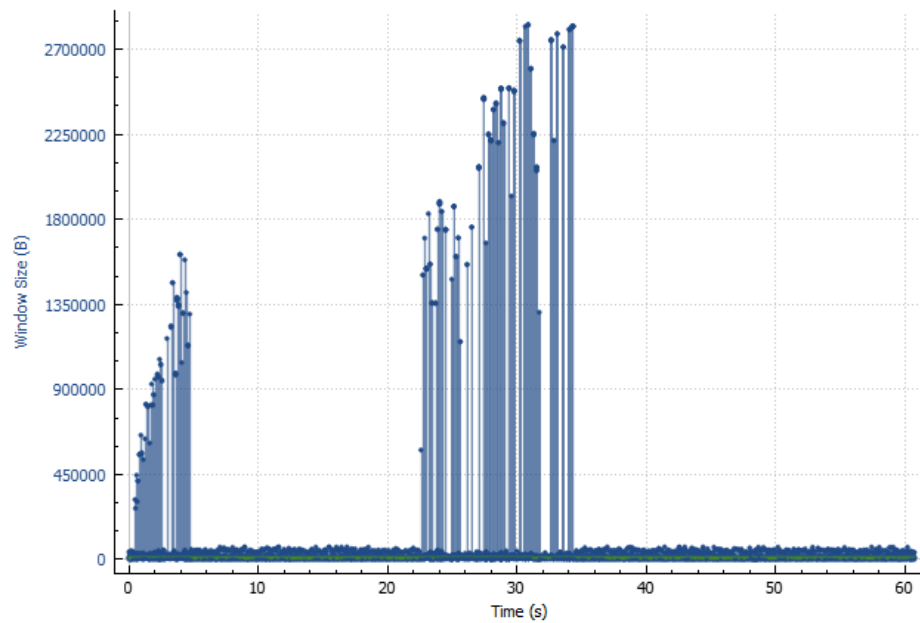


Figure 30: Congestion SVI (EVE-NG)

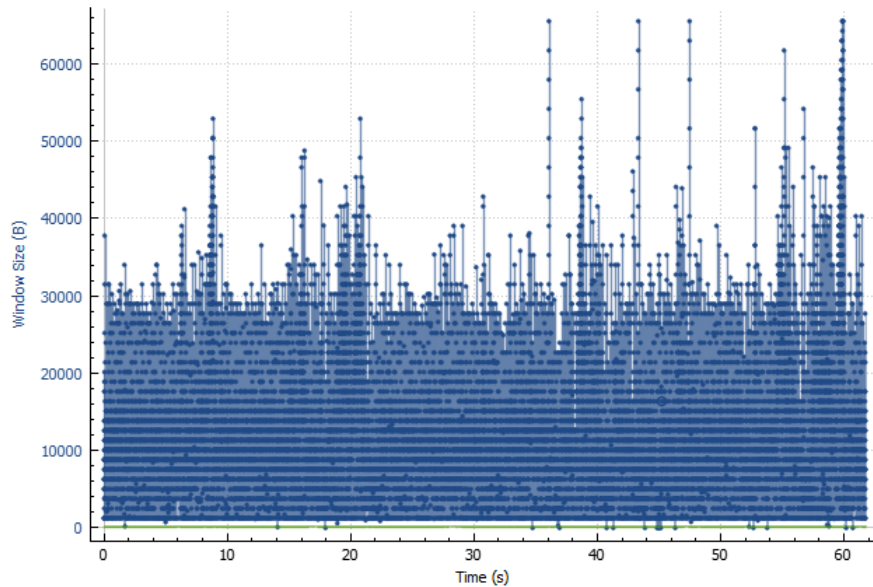


Figure 31: Congestion SVI (GNS3)

4.2.5 Scalability

Its capacity to manage escalating resource needs and accept expanding traffic or user loads without significantly degrading performance entails creating a network architecture and adopting technologies that are easily adaptable and expandable to accommodate more users, devices, data quantities, or transactions [33].

Scalability ON EVE-ng

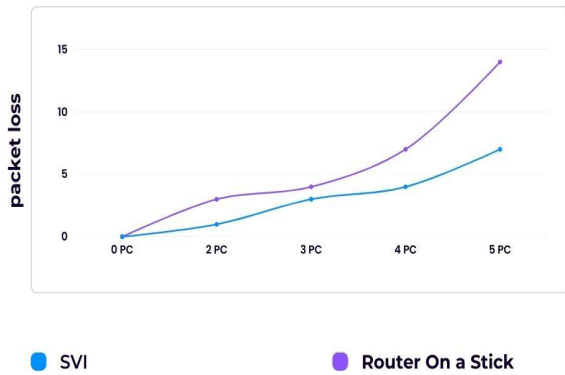


Figure 32 : Scalability (EVE-NG)

Scalability ON GNS3

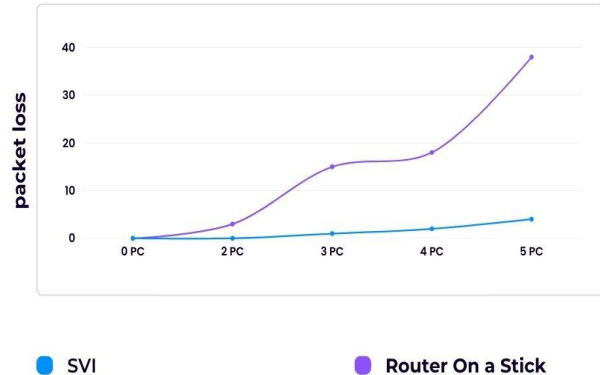


Figure 33 : Scalability (GNS3)

According to Figures 32, 33, Switched Virtual Interface (SVI) is a more scalable method for inter-VLAN communication than Router on a Stick (ROAS). Our research indicated that SVI is better able to manage a larger number of VLANs with more efficiency since it allows inter-VLAN traffic to be handled within the switch itself. In contrast, ROAS relies on a single physical interface to manage all inter-VLAN traffic, limiting scalability and potentially causing network congestion and bottlenecks as the number of VLANs grows. According to our findings, SVI gives network managers more freedom in VLAN administration, allowing them to add and delete VLANs as needed without severely compromising network performance. As a result of our findings, we advocate SVI as the more scalable alternative in network architecture for inter-VLAN communication [33].

4.2.6 Reliability

According to the results above, we can say SVI is more reliable than a router on a stick. There are various things to consider when it comes to reliability, including performance, scalability, and adaptability. Based on the findings of various assessments or tests, it is possible to infer that SVI is more dependable than Router on a stick in certain areas. SVI, for example, can outperform a router on a stick in scenarios with a high number of VLANs because it eliminates the bottleneck that might emerge when all traffic must pass via a single physical interface on the router. Moreover, SVI is more scalable than Router on a Stick since it permits the formation of many virtual interfaces on the switch, whereas Router on a Stick has a restriction on the number of VLANs that can be handled by a single physical interface. Additionally, SVI provides greater flexibility since it allows inter-VLAN routing to be conducted on the switch itself rather than on

the router. This can minimize network complexity, enhance network architecture, and facilitate troubleshooting. Generally, keep in mind that the dependability of SVI against RoAS will be determined by the unique needs of a given network. While one strategy may be more acceptable in some cases, the other may be more suitable in others. As a result, it is critical to thoroughly assess each alternative and select the one that best fulfills the network's objectives [34].

4.2.7 Cost

Our study compared the cost and price of installing layer 3 switching vs. the use of routers in network architecture. Layer 3 switching uses a switch that can also perform routing duties, whereas routers are specialized networking equipment developed solely for routing packets across various networks. After thorough cost and price research, we discovered that layer 3 switching is often a more cost-effective and economical alternative than routers in many network architecture scenarios. Layer 3 switches often cost less than routers and can provide quicker packet processing and more port density, resulting in cheaper operational expenses. However, it is crucial to remember that the decision between layer 3 switching and routers is dependent on the network's individual demands and requirements. In some circumstances, routers may still be the preferable solution, particularly for networks requiring complicated routing protocols or substantial network segmentation. Finally, the selection should be based on a thorough examination of the network's requirements and available possibilities [12] [22].

4.2.8 Security

We'll look at the possible security concerns and threats that might occur while deploying inter-VLAN routing, as well as tactics and best practices for addressing them. Network administrators may make educated judgments regarding network design and architecture, as well as assure the security and protection of their networks, by knowing the security implications of inter-VLAN routing. [32] Without understanding the particular nature of the assault and the security features and setups of each device, it's difficult to predict which device would fight better against an attack. Both routers and switches may be targets for attackers, and both can have exploitable weaknesses [18] [32]. Routers on the other hand, are often constructed with more complex security capabilities and are frequently regarded as the first line of defense in a network. Granular control over traffic flow and routing policies, enhanced firewalling and intrusion protection, and more flexible interaction with other devices and protocols can all be provided by routers. These characteristics can make it more difficult for attackers to exploit vulnerabilities or get network access [32]. Switches, on the other hand, are primarily intended for network connections and often lack the comprehensive security capabilities found in routers. While switches may implement VLAN-based access control and

other basic security measures, they may fall short when it comes to fighting against more sophisticated assaults. To minimize the danger of attacks, both routers and switches should be set up with proper security measures and periodically updated with security updates. It's also critical to monitor network activity and have incident response strategies in place to detect and respond to any security problems as soon as they occur.

4.2.8.1 Access Control List (ACL)

ACLs are a form of security mechanism that is used in routers and Layer 3 switches to regulate network access. ACLs enable network managers to accept or restrict specific types of traffic based on parameters like IP address, port number, and protocol. Network administrators can prevent attackers from getting access to the network by installing an ACL that allows specified IP addresses but prohibits others. For example, if an attacker attempts to connect to the network using an IP address that has been rejected in the ACL, the router or Layer 3 switch will block the traffic and prevent the attacker from connecting to the network [34]. This security technique is very useful since it allows network managers to regulate network access at a granular level. Network administrators can verify that only permitted traffic enters the network by designating which IP addresses are allowed or banned. Potentially harmful traffic is prevented. Using ACLs to allow specific IP addresses and reject others is an essential security strategy that may help protect networks from possible threats and guarantee that only permitted traffic passes through the network [35] [36].

```
root@kali:~# ping 192.168.13.1
PING 192.168.13.1 (192.168.13.1) 56(84) bytes of data.
64 bytes from 192.168.13.1: icmp_seq=1 ttl=127 time=14.3 ms
64 bytes from 192.168.13.1: icmp_seq=2 ttl=127 time=20.5 ms
64 bytes from 192.168.13.1: icmp_seq=3 ttl=127 time=15.6 ms
64 bytes from 192.168.13.1: icmp_seq=4 ttl=127 time=14.7 ms
64 bytes from 192.168.13.1: icmp_seq=5 ttl=127 time=13.8 ms
64 bytes from 192.168.13.1: icmp_seq=6 ttl=127 time=18.4 ms
64 bytes from 192.168.13.1: icmp_seq=7 ttl=127 time=17.3 ms
■
```

Figure 34 : Attacker PC ping to PC Vlan 13 (Before ACL)

Figure 34 shows that, prior to setting the Access Control List (ACL), an attacker can launch a ping command against other machines.

```
Router(config)#access-list 3 permit host 192.168.11.1
Router(config)#access-list 3 permit host 192.168.12.1
Router(config)#access-list 3 permit host 192.168.13.1
Router(config)#access-list 3 permit host 192.168.14.1
Router(config)#access-list 3 permit host 192.168.15.2
Router(config)#
Router(config)#acc
Router(config)#access-list 3 deny any
Router(config)#
```

Figure 35 :Implementing ACL (Router)

In Figure 35, it is evident that we have augmented the Access Control List (ACL) by incorporating a trusted IP address.

```

round trip min/avg/max = 0.0/0.0/0.0 ms
root@kali:~# ping 192.168.13.1
PING 192.168.13.1 (192.168.13.1) 56(84) bytes of data.
From 192.168.14.14 icmp_seq=1 Packet filtered
From 192.168.14.14 icmp_seq=2 Packet filtered
From 192.168.14.14 icmp_seq=3 Packet filtered
From 192.168.14.14 icmp_seq=4 Packet filtered
From 192.168.14.14 icmp_seq=5 Packet filtered
From 192.168.14.14 icmp_seq=6 Packet filtered
From 192.168.14.14 icmp_seq=7 Packet filtered
From 192.168.14.14 icmp_seq=8 Packet filtered

```

Figure 36 : Attacker PC ping to PC Vlan 13 (After ACL)

In Figure 36 , When implemented in routers or Layer 3 switches, access control lists (ACLs) can prohibit attackers from transmitting packets to different VLANs inside a network. This is due to the fact that ACLs can designate which IP addresses are permitted to connect with one another and which are not. If an attacker attempts to transmit packets to VLANs that are not inside the designated boundary, the ACL will reject the packets and prohibit the attacker from connecting with those VLANs. This implies that only approved traffic is permitted to transit via the network, while potentially harmful traffic is denied. Establishing ACLs improves network security and protects against potential threats. Network managers can safeguard their network from unwanted access and potential security issues by selecting which IP addresses are allowed to connect with one another.

4.2.8.2 Attack with Kali Linux

4.2.8.2.1 TCP Attack Hping3

Hping3 is a flexible network tool that, like the traditional ping utility, allows users to send customized ICMP, UDP, or TCP packets and examine the answers from the target device. It enables file transfers across supported protocols by fragmenting packets and adjusting their size and contents. The program is especially useful for testing firewall rules, doing spoofed port scans, and assessing network performance across many protocols. Among other things, Hping3 may be used to find the path MTU, execute traceroute-like actions across several protocols, identify remote OS systems, and audit TCP/IP stacks. In addition, the software may be created in the TCL programming language, which increases its versatility and simplicity of customization. Overall, hping3 is a must-have application for network administrators and security experts, offering a plethora of functions for testing and enhancing network security and performance [37]. In figures 37 , 38 , 39 . Attack from Kali 1 in Vlan 14 to PCs in Vlan 11,12,13.

```
root@kali: ~ x root@kali: ~ x root@kali: ~ x root@kali: ~ x
.5 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8830 sport=0 flags=RA seq=8608 win=0 rtt=19
.9 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8831 sport=0 flags=RA seq=8609 win=0 rtt=18
.9 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8832 sport=0 flags=RA seq=8610 win=0 rtt=60
.9 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8833 sport=0 flags=RA seq=8611 win=0 rtt=49
.5 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8834 sport=0 flags=RA seq=8612 win=0 rtt=41
.2 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8835 sport=0 flags=RA seq=8613 win=0 rtt=29
.9 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8836 sport=0 flags=RA seq=8614 win=0 rtt=17
.6 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8837 sport=0 flags=RA seq=8615 win=0 rtt=33
.8 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8838 sport=0 flags=RA seq=8616 win=0 rtt=23
.1 ms
len=46 ip=192.168.11.1 ttl=127 DF id=8839 sport=0 flags=RA seq=8617 win=0 rtt=21
.7 ms
```

Figure 37 : Hping3 attack to PC Vlan 11

```
root@kali: ~ x root@kali: ~ x root@kali: ~ x root@kali: ~ x
.0 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7545 sport=0 flags=RA seq=7031 win=0 rtt=26
.9 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7546 sport=0 flags=RA seq=7032 win=0 rtt=19
.8 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7547 sport=0 flags=RA seq=7033 win=0 rtt=17
.0 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7548 sport=0 flags=RA seq=7034 win=0 rtt=25
.8 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7549 sport=0 flags=RA seq=7035 win=0 rtt=23
.6 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7550 sport=0 flags=RA seq=7036 win=0 rtt=18
.6 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7551 sport=0 flags=RA seq=7037 win=0 rtt=25
.2 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7552 sport=0 flags=RA seq=7038 win=0 rtt=20
.9 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7553 sport=0 flags=RA seq=7039 win=0 rtt=18
.0 ms
len=46 ip=192.168.12.1 ttl=127 DF id=7554 sport=0 flags=RA seq=7040 win=0 rtt=14
.7 ms

root@kali: ~ x root@kali: ~ x root@kali: ~ x root@kali: ~ x
.8 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6435 sport=0 flags=RA seq=5199 win=0 rtt=27
.8 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6436 sport=0 flags=RA seq=5200 win=0 rtt=21
.7 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6437 sport=0 flags=RA seq=5201 win=0 rtt=19
.4 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6438 sport=0 flags=RA seq=5202 win=0 rtt=18
.1 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6439 sport=0 flags=RA seq=5203 win=0 rtt=31
.0 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6440 sport=0 flags=RA seq=5204 win=0 rtt=28
.0 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6441 sport=0 flags=RA seq=5205 win=0 rtt=25
.9 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6442 sport=0 flags=RA seq=5206 win=0 rtt=23
.3 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6443 sport=0 flags=RA seq=5207 win=0 rtt=20
.7 ms
len=46 ip=192.168.13.1 ttl=127 DF id=6444 sport=0 flags=RA seq=5208 win=0 rtt=18
.6 ms
```

Figure 38 : Hping3 attack to PC Vlan 12

Figure 39 : Hping3 attack to PC Vlan 13

Figures 40 to 47 show detailed data and graphical representations of the Hping3 attack obtained in Wireshark. According to the data shown, the bulk of the assaults explicitly target the switch. The main reason for the switch's concentration of attacks is its capacity to efficiently handle and manage a larger volume of network traffic than other devices.

No.	Time	Source	Destination	Protocol	Length	Info
12464	11.307893	192.168.14.2	192.168.13.1	TCP	64	12986 → 0 [None] Seq=1 Win=512 Len=0
12465	11.309402	192.168.14.2	192.168.12.1	TCP	64	14844 → 0 [None] Seq=1 Win=512 Len=0
12466	11.310104	192.168.14.2	192.168.11.1	TCP	64	14240 → 0 [None] Seq=1 Win=512 Len=0
12467	11.312457	192.168.12.1	192.168.14.2	TCP	58	0 → 14843 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12468	11.312501	192.168.11.1	192.168.14.2	TCP	58	0 → 14239 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12469	11.312507	192.168.13.1	192.168.14.2	TCP	58	0 → 12985 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12470	11.312518	192.168.14.2	192.168.13.1	TCP	58	[TCP Dup ACK 12464#1] 12986 → 0 [None] Seq=1 Win=512 Len=0
12471	11.312520	192.168.14.2	192.168.12.1	TCP	58	[TCP Dup ACK 12465#1] 14844 → 0 [None] Seq=1 Win=512 Len=0
12472	11.312522	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 12466#1] 14240 → 0 [None] Seq=1 Win=512 Len=0
12473	11.314064	192.168.11.1	192.168.14.2	TCP	58	0 → 14240 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12474	11.314083	192.168.12.1	192.168.14.2	TCP	58	0 → 14844 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12475	11.314219	192.168.13.1	192.168.14.2	TCP	58	0 → 12986 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12476	11.318480	192.168.14.2	192.168.13.1	TCP	64	12987 → 0 [None] Seq=1 Win=512 Len=0
12477	11.320446	192.168.14.2	192.168.12.1	TCP	64	14845 → 0 [None] Seq=1 Win=512 Len=0
12478	11.321194	192.168.14.2	192.168.11.1	TCP	64	14241 → 0 [None] Seq=1 Win=512 Len=0
12479	11.322816	192.168.11.1	192.168.14.2	TCP	58	0 → 14240 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12480	11.322845	192.168.12.1	192.168.14.2	TCP	58	0 → 14844 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12481	11.322850	192.168.13.1	192.168.14.2	TCP	58	0 → 12986 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12482	11.322852	192.168.14.2	192.168.13.1	TCP	58	[TCP Dup ACK 12476#1] 12987 → 0 [None] Seq=1 Win=512 Len=0
12483	11.322854	192.168.14.2	192.168.12.1	TCP	58	[TCP Dup ACK 12477#1] 14845 → 0 [None] Seq=1 Win=512 Len=0
12484	11.322856	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 12478#1] 14241 → 0 [None] Seq=1 Win=512 Len=0
12485	11.324301	192.168.11.1	192.168.14.2	TCP	58	0 → 14241 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12486	11.324323	192.168.12.1	192.168.14.2	TCP	58	0 → 14845 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12487	11.324327	192.168.13.1	192.168.14.2	TCP	58	0 → 12987 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
12488	11.329113	192.168.14.2	192.168.13.1	TCP	64	12988 → 0 [None] Seq=1 Win=512 Len=0
12489	11.331486	192.168.14.2	192.168.12.1	TCP	64	14846 → 0 [None] Seq=1 Win=512 Len=0
12490	11.331508	192.168.14.2	192.168.11.1	TCP	64	14242 → 0 [None] Seq=1 Win=512 Len=0

Figure 40 : Hping3 attack information on wireshark (Router In GNS3)

No.	Time	Source	Destination	Protocol	Length	Info
4746	4.343289	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 4745#1] 3428 → 0 [None] Seq=1 Win=512 Len=0
4747	4.343880	192.168.11.1	192.168.14.2	TCP	58	0 → 3428 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4748	4.343919	192.168.11.1	192.168.14.2	TCP	58	0 → 3428 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4749	4.349824	192.168.14.2	192.168.13.1	TCP	64	2594 → 0 [None] Seq=1 Win=512 Len=0
4750	4.349996	192.168.14.2	192.168.13.1	TCP	58	[TCP Dup ACK 4749#1] 2594 → 0 [None] Seq=1 Win=512 Len=0
4751	4.350093	192.168.14.2	192.168.12.1	TCP	64	3584 → 0 [None] Seq=1 Win=512 Len=0
4752	4.350131	192.168.14.2	192.168.12.1	TCP	58	[TCP Dup ACK 4751#1] 3584 → 0 [None] Seq=1 Win=512 Len=0
4753	4.351012	192.168.13.1	192.168.14.2	TCP	58	0 → 2594 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4754	4.351059	192.168.13.1	192.168.14.2	TCP	58	0 → 2594 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4755	4.351256	192.168.12.1	192.168.14.2	TCP	58	0 → 3584 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4756	4.351283	192.168.12.1	192.168.14.2	TCP	58	0 → 3584 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4757	4.353736	192.168.14.2	192.168.11.1	TCP	64	3429 → 0 [None] Seq=1 Win=512 Len=0
4758	4.353786	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 4757#1] 3429 → 0 [None] Seq=1 Win=512 Len=0
4759	4.354303	192.168.11.1	192.168.14.2	TCP	58	0 → 3429 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4760	4.354344	192.168.11.1	192.168.14.2	TCP	58	0 → 3429 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4761	4.360189	192.168.14.2	192.168.13.1	TCP	64	2595 → 0 [None] Seq=1 Win=512 Len=0
4762	4.360248	192.168.14.2	192.168.13.1	TCP	58	[TCP Dup ACK 4761#1] 2595 → 0 [None] Seq=1 Win=512 Len=0
4763	4.360562	192.168.14.2	192.168.12.1	TCP	64	3585 → 0 [None] Seq=1 Win=512 Len=0
4764	4.360592	192.168.14.2	192.168.12.1	TCP	58	[TCP Dup ACK 4763#1] 3585 → 0 [None] Seq=1 Win=512 Len=0
4765	4.361285	192.168.13.1	192.168.14.2	TCP	58	0 → 2595 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4766	4.361344	192.168.13.1	192.168.14.2	TCP	58	0 → 2595 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4767	4.361589	192.168.12.1	192.168.14.2	TCP	58	0 → 3585 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4768	4.361637	192.168.12.1	192.168.14.2	TCP	58	0 → 3585 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4769	4.364139	192.168.14.2	192.168.11.1	TCP	64	3430 → 0 [None] Seq=1 Win=512 Len=0
4770	4.364191	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 4769#1] 3430 → 0 [None] Seq=1 Win=512 Len=0
4771	4.364731	192.168.11.1	192.168.14.2	TCP	58	0 → 3430 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
4772	4.364772	192.168.11.1	192.168.14.2	TCP	58	0 → 3430 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Figure 41 : Hping3 attack information on wireshark (Switch In GNS3)

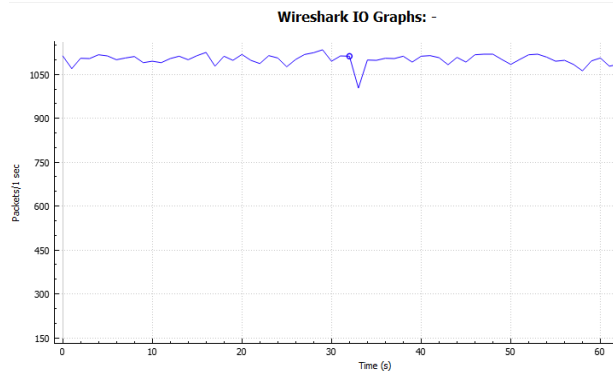


Figure 42 : Hping3 Graphs(Router In GNS3)

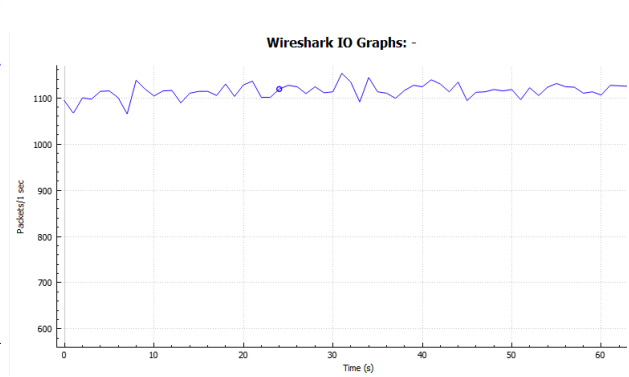


Figure 43 : Hping3 Graphs (Switch In GNS3)

No.	Time	Source	Destination	Protocol	Length	Info
98421	84.920397	192.168.13.1	192.168.14.2	TCP	58	0 → 11100 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98422	84.920465	192.168.12.1	192.168.14.2	TCP	58	0 → 12453 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98423	84.920516	192.168.14.2	192.168.11.1	TCP	58	12950 → 0 [<None>] Seq=1 Win=512 Len=0
98424	84.924984	192.168.14.2	192.168.12.1	TCP	58	12454 → 0 [<None>] Seq=1 Win=512 Len=0
98425	84.925008	192.168.14.2	192.168.13.1	TCP	58	11101 → 0 [<None>] Seq=1 Win=512 Len=0
98426	84.929659	192.168.11.1	192.168.14.2	TCP	58	0 → 12949 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98427	84.929680	192.168.13.1	192.168.14.2	TCP	58	0 → 11100 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98428	84.929683	192.168.12.1	192.168.14.2	TCP	58	0 → 12453 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98429	84.929684	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 98423#1] 12950 → 0 [<None>] Seq=1 Win=51...
98430	84.929684	192.168.14.2	192.168.12.1	TCP	58	[TCP Dup ACK 98424#1] 12454 → 0 [<None>] Seq=1 Win=51...
98431	84.929686	192.168.14.2	192.168.13.1	TCP	58	[TCP Dup ACK 98425#1] 11101 → 0 [<None>] Seq=1 Win=51...
98432	84.930387	192.168.11.1	192.168.14.2	TCP	58	0 → 12950 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98433	84.930774	192.168.12.1	192.168.14.2	TCP	58	0 → 12454 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98434	84.930798	192.168.13.1	192.168.14.2	TCP	58	0 → 11101 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98435	84.930982	192.168.14.2	192.168.11.1	TCP	58	12951 → 0 [<None>] Seq=1 Win=512 Len=0
98436	84.935469	192.168.14.2	192.168.13.1	TCP	58	11102 → 0 [<None>] Seq=1 Win=512 Len=0
98437	84.935495	192.168.14.2	192.168.12.1	TCP	58	12455 → 0 [<None>] Seq=1 Win=512 Len=0
98438	84.940065	192.168.11.1	192.168.14.2	TCP	58	0 → 12950 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98439	84.940082	192.168.12.1	192.168.14.2	TCP	58	0 → 12454 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98440	84.940084	192.168.13.1	192.168.14.2	TCP	58	0 → 11101 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98441	84.940085	192.168.14.2	192.168.11.1	TCP	58	[TCP Dup ACK 98435#1] 12951 → 0 [<None>] Seq=1 Win=51...
98442	84.940085	192.168.14.2	192.168.13.1	TCP	58	[TCP Dup ACK 98436#1] 11102 → 0 [<None>] Seq=1 Win=51...
98443	84.940087	192.168.14.2	192.168.12.1	TCP	58	[TCP Dup ACK 98437#1] 12455 → 0 [<None>] Seq=1 Win=51...
98444	84.940783	192.168.11.1	192.168.14.2	TCP	58	0 → 12951 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98445	84.940884	192.168.13.1	192.168.14.2	TCP	58	0 → 11102 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98446	84.941164	192.168.12.1	192.168.14.2	TCP	58	0 → 12455 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
98447	84.941342	192.168.14.2	192.168.11.1	TCP	58	12952 → 0 [<None>] Seq=1 Win=512 Len=0

Figure 44 : Hping3 attack information on wireshark (Router In EVE-NG)

No.	Time	Source	Destination	Protocol	Length	Info
37949	32.708608	192.168.14.11	192.168.11.1	TCP	58	[TCP Dup ACK 37948#1] 7650 → 0 [<None>] Seq=1 Win=512...
37950	32.709396	192.168.11.1	192.168.14.11	TCP	58	0 → 7650 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37951	32.709498	192.168.11.1	192.168.14.11	TCP	58	0 → 7650 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37952	32.711549	192.168.14.11	192.168.13.1	TCP	58	7862 → 0 [<None>] Seq=1 Win=512 Len=0
37953	32.711648	192.168.14.11	192.168.13.1	TCP	58	[TCP Dup ACK 37952#1] 7862 → 0 [<None>] Seq=1 Win=512...
37954	32.712379	192.168.13.1	192.168.14.11	TCP	58	0 → 7862 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37955	32.712466	192.168.13.1	192.168.14.11	TCP	58	0 → 7862 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37956	32.715920	192.168.14.11	192.168.12.1	TCP	58	6989 → 0 [<None>] Seq=1 Win=512 Len=0
37957	32.716070	192.168.14.11	192.168.12.1	TCP	58	[TCP Dup ACK 37956#1] 6989 → 0 [<None>] Seq=1 Win=512...
37958	32.716780	192.168.12.1	192.168.14.11	TCP	58	0 → 6989 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37959	32.716864	192.168.12.1	192.168.14.11	TCP	58	0 → 6989 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37960	32.719889	192.168.14.11	192.168.11.1	TCP	58	7651 → 0 [<None>] Seq=1 Win=512 Len=0
37961	32.720005	192.168.14.11	192.168.11.1	TCP	58	[TCP Dup ACK 37960#1] 7651 → 0 [<None>] Seq=1 Win=512...
37962	32.720796	192.168.11.1	192.168.14.11	TCP	58	0 → 7651 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37963	32.720872	192.168.11.1	192.168.14.11	TCP	58	0 → 7651 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37964	32.721809	192.168.14.11	192.168.13.1	TCP	58	7863 → 0 [<None>] Seq=1 Win=512 Len=0
37965	32.721870	192.168.14.11	192.168.13.1	TCP	58	[TCP Dup ACK 37964#1] 7863 → 0 [<None>] Seq=1 Win=512...
37966	32.722627	192.168.13.1	192.168.14.11	TCP	58	0 → 7863 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37967	32.722715	192.168.13.1	192.168.14.11	TCP	58	0 → 7863 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37968	32.725595	192.168.14.11	192.168.12.1	TCP	58	6990 → 0 [<None>] Seq=1 Win=512 Len=0
37969	32.725670	192.168.14.11	192.168.12.1	TCP	58	[TCP Dup ACK 37968#1] 6990 → 0 [<None>] Seq=1 Win=512...
37970	32.726296	192.168.12.1	192.168.14.11	TCP	58	0 → 6990 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37971	32.726389	192.168.12.1	192.168.14.11	TCP	58	0 → 6990 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37972	32.730592	192.168.14.11	192.168.11.1	TCP	58	7652 → 0 [<None>] Seq=1 Win=512 Len=0
37973	32.730717	192.168.14.11	192.168.11.1	TCP	58	[TCP Dup ACK 37972#1] 7652 → 0 [<None>] Seq=1 Win=512...
37974	32.731489	192.168.11.1	192.168.14.11	TCP	58	0 → 7652 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
37975	32.731601	192.168.11.1	192.168.14.11	TCP	58	0 → 7652 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Figure 45 : Hping3 attack information on wireshark (Switch In EVE-NG)

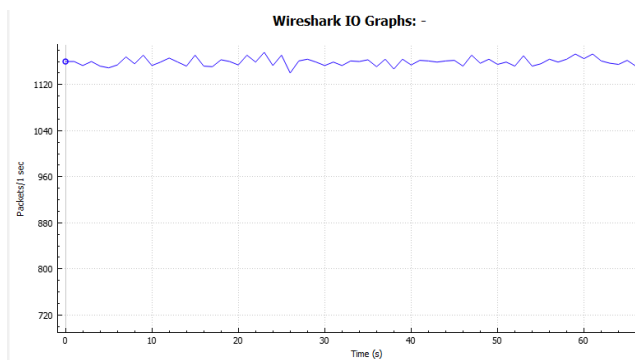


Figure 46 : Hping3 Graphs (Router In EVE-NG)

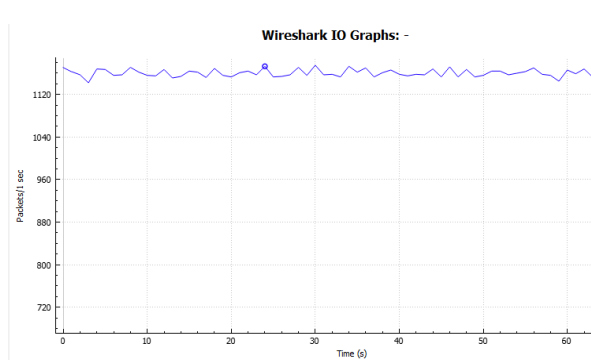


Figure 47 : Hping3 Graphs (Switch In EVE-NG)

4.2.8.2.1 CDP Attack with Yersinia

Yersinia is a strong framework that specializes in layer 2 attacks that take advantage of flaws in various network protocols. It positions itself as a necessary tool for studying and testing network systems and infrastructure. Its capabilities go beyond conventional penetration testing, allowing for a more in-depth knowledge of network security concerns by finding vulnerabilities that might be exploited by malevolent actors. The program is intended to mimic real-world network assaults, giving network managers a full picture of their network's security posture. It is an excellent tool for finding security flaws and vulnerabilities that attackers may exploit, making it an essential component of any security assessment. For network administrators, security specialists, and enterprises that emphasize network security, Yersinia is a crucial tool. Its extensive capabilities make it a vital tool for identifying and mitigating network threats and

improving network security. [38] When it comes to network security, routers outperform switches. This is because router ports are always in a shutdown state and do not have an IP address assigned to them. As a result, carrying out Yersinia CDP assaults on the network is significantly more difficult. This is a crucial concern when adopting network security measures since Yersinia is a formidable instrument for carrying out layer 2 assaults that might jeopardize network security. Organizations may dramatically lower the possibility of such attacks and improve their overall network security posture by using a router with default shutdown ports and no IP address. [38]

The Result is After 60 second .attacker is kali2 to router and switch commands:

show cdp neighbors to know the attack is working [39].

show processes cpu sorted | include CPU | PID Runtime | CDP Protocol for the usage of cpu in router and switch that yersinia uses for the attack [39].

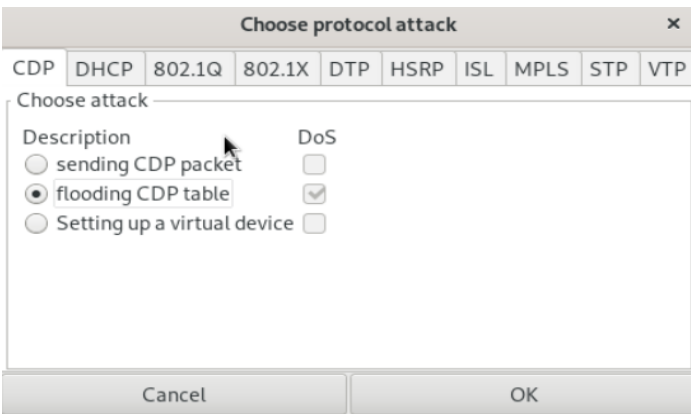


Figure 48 :sending CDP attack with yersinia

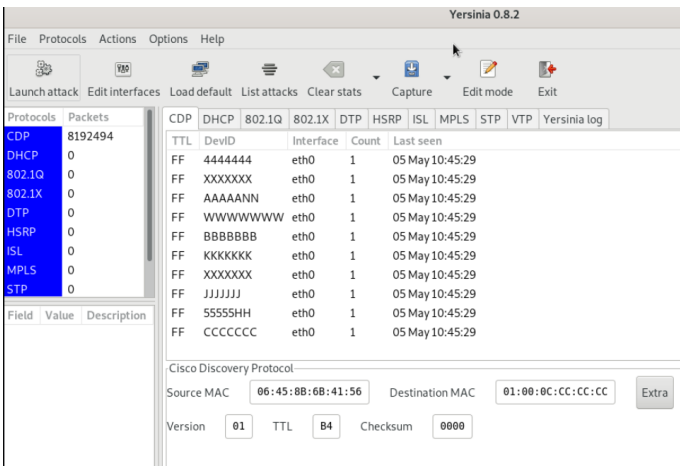


Figure 49 : CDP attack with yersinia

The technique represented in Figures 48 and 49 illustrates the procedures necessary in launching a CDP attack against the connected port, regardless of whether it is a switch or a router. Figure 50 also depicts the continued execution of the Yersinia assault. Figures 51, 52, 53, and 54 show the CPU performance of the switch and router in both simulations.

```
Router#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability Platform Port ID
00000000          Eth 4/0         186        S I r   yersinia Eth 0
00000000          Eth 4/0         186        R T I   yersinia Eth 0
00000000          Eth 4/0         186        R T S H I yersinia Eth 0
00000000          Eth 4/0         186        R T I   yersinia Eth 0
000RRRRR          Eth 4/0         186        R S I r  yersinia Eth 0
00000000          Eth 4/0         186        R T S I  yersinia Eth 0
SS000000          Eth 4/0         186        R T S H I yersinia Eth 0
00000000          Eth 4/0         186        R T S I  yersinia Eth 0
00000000          Eth 4/0         186        R T I   yersinia Eth 0
00000000          Eth 4/0         186        R T S I  yersinia Eth 0
00000000          Eth 4/0         186        R T H I  yersinia Eth 0
RRRRRRR0          Eth 4/0         186        R T B H  yersinia Eth 0
00000000          Eth 4/0         186        R T I   yersinia Eth 0
00000000          Eth 4/0         186        R T S H I yersinia Eth 0
00000000          Eth 4/0         185        R T S I  yersinia Eth 0
00000000          Eth 4/0         185        R T B S  yersinia Eth 0
VVVV0000          Eth 4/0         185        R T S I  yersinia Eth 0
00000000          Eth 4/0         185        R T H I  yersinia Eth 0
00000000          Eth 4/0         185        R T B H  yersinia Eth 0
00000000          Eth 4/0         185        R T S I  yersinia Eth 0
--More--
```

Figure 50 :CDP Neighbours

```
Router#
Router#show processes cpu sorted | include CPU | PID Runtime | CDP Protocol
CPU utilization for five seconds: 99%/100%; one minute: 63%; five minutes: 18%
 95      61248      4466      13714 81.86% 47.55% 13.63%  0 CDP Protocol
Router#
```

Figure 51 : processes cpu sorted include CPU PID Runtime CDP Protocol (Router in GNS3)

```
SW8#
SW8#show processes cpu sorted | include CPU | PID Runtime | CDP Protocol
CPU utilization for five seconds: 0%/0%; one minute: 0%; five minutes: 0%
 92      322      578578      0 35.67% 21.60% 6.09%  0 CDP Protocol
SW8#
```

Figure 52 : processes cpu sorted include CPU PID Runtime CDP Protocol (Switch in GNS3)


```
Router#show processes cpu sorted | include CPU | PID Runtime | CDP Protocol
CPU utilization for five seconds: 99%/100%; one minute: 64%; five minutes: 18%
 96      61412      5367      11442 92.23% 57.70% 16.59%  0 CDP Protocol
Router#
```

Figure 53 : processes cpu sorted include CPU PID Runtime CDP Protocol (Router in EVE-NG)

```
Switch#show processes cpu sorted | include CPU | PID Runtime | CDP Protocol
CPU utilization for five seconds: 71%/0%; one minute: 31%; five minutes: 7%
 92      37810      396748      95 100.00% 59.79% 16.75%  0 CDP Protocol
Switch#
```

Figure 54 : processes cpu sorted include CPU PID Runtime CDP Protocol (Switch in EVE-NG)

No.	Time	Source	Destination	Protocol	Length	Info
3428...	15.480907	e9:c7:92:26:f3:23	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 8888888 Port ID: Ethernet0
3428...	15.480983	92:19:f0:65:7f:99	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: JJJJJJJ Port ID: Ethernet0
3428...	15.481017	b3:81:ba:25:41:9e	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: NN66666 Port ID: Ethernet0
3428...	15.481054	1d:d7:e2:1c:a7:eb	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: DDDDDDD Port ID: Ethernet0
3428...	15.481139	9b:26:65:4d:ca:3a	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 8888888 Port ID: Ethernet0
3428...	15.481163	51:23:06:39:e6:04	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: TTTTTTT Port ID: Ethernet0
3428...	15.481188	de:09:ed:21:e7:a4	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 8888888 Port ID: Ethernet0
3428...	15.481264	73:3b:a7:52:48:32	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 22222II Port ID: Ethernet0
3428...	15.481288	c9:3c:09:1e:10:92	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: MTTTTTTT Port ID: Ethernet0
3428...	15.481313	51:78:5d:4c:3c:15	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 55IIIII Port ID: Ethernet0
3428...	15.481340	8b:6b:e8:71:b8:03	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 4444444 Port ID: Ethernet0
3428...	15.481343	fa:40:4a:74:3b:03	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: L444444 Port ID: Ethernet0
3428...	15.481366	3d:50:11:12:a8:aa	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: QQQQQQQ Port ID: Ethernet0
3428...	15.481439	3b:e3:f6:0c:a8:1e	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 7777777 Port ID: Ethernet0
3429...	15.481463	b5:b9:de:44:a8:92	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: A000000 Port ID: Ethernet0
3429...	15.481527	79:1f:6b:67:ff:6e	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 0000000 Port ID: Ethernet0
3429...	15.481605	29:29:80:1c:ca:9a	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: VVVVVVV Port ID: Ethernet0
3429...	15.481629	5f:02:e0:15:e4:a8	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: ZGGGGGG Port ID: Ethernet0
3429...	15.481653	c4:54:f9:27:b8:03	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: YYYYYYY Port ID: Ethernet0
3429...	15.481678	15:e0:6b:4d:66:c8	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: GGGGGGG Port ID: Ethernet0
3429...	15.481754	ab:78:43:4e:da:25	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 7777777 Port ID: Ethernet0
3429...	15.481778	e4:fc:6a:07:1e:5b	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: NNNNNNN Port ID: Ethernet0
3429...	15.481802	63:d7:cb:03:6c:bd	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 0000000 Port ID: Ethernet0
3429...	15.481827	fb:e3:54:6c:9e:ad	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: 5555555 Port ID: Ethernet0
3429...	15.481902	3c:64:dc:4c:32:2c	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: GGGGGGG Port ID: Ethernet0
3429...	15.481945	31:01:7a:4b:17:33	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: GGGGGGG Port ID: Ethernet0
3429...	15.481975	f2:50:32:1c:c0:78	CDP/VTP/DTP/PagP/UD...	CDP	96	Device ID: FFFFFFF Port ID: Ethernet0

Figure 55 :CDP attack (wireshark)

Name	CPU usage percentage % for one minute
Router in GNS3	63
Router in EVE-NG	64
Switch in GNS3	0 the attack not fully working on switch in GNS3
Switch in EVE-NG	31

Table 6 :CPU usage after CDP attack

Following the assault, the Table 6 analysis indicated that the router's CPU utilization was higher than the switch layer 3's. This discrepancy, however, was only visible while the router port was active. It is important to notice that the switch's layer 3 functionality runs continually, ensuring that it is always available and active. As a consequence, because the conditions under which the two devices were evaluated were not exactly equal, it is not feasible to proclaim the switch better purely based on the findings of CPU utilization [39].

4.2.8.3 Port security

When it comes to port security, routers outperform switches. This is because router ports are always in a shutdown state, but switch ports are always active and must be manually turned down using a command. This router default option adds an extra layer of protection to the network by preventing unwanted access and network devices from connecting to it. This is especially crucial for firms that deal with sensitive data or need to implement strict security procedures. Switches, on the other hand, need a more manual approach to port security, which may be time-consuming and prone to human error. Administrators must manually shut down each port that is not in use, which can be a time-consuming operation that may result in some ports being left open accidentally [40]. Because their default settings give a better level of protection and limit the possibility of unwanted access or security breaches, routers are a more efficient and secure alternative for port security.

5. Evaluation

This study Focused at inter-vlan routing in a virtual network environment using Switch Virtual Interfaces (SVI) and Router on a Stick (RoAS) configurations. One of the study's main strengths is that it fills a gap in the literature by comparing these two techniques. Furthermore, this study compares the two settings using a comprehensive set of network measures, which is an improvement above prior studies, which sometimes used just two or three indicators. Another highlight of this study is the use of a powerful PC to mimic network architecture and virtual devices. This method gives a realistic picture of real-world outcomes, which increases the credibility of the study's conclusions. Furthermore, using a fully functional FTP server to generate network traffic is a significant advance above previous experiments that used simpler protocols like FTP or ping. While some studies use FTP, they do not use it as extensively as in the real world [15] [16], and they also employ other technologies such as HTTP, DATABASE [17], ping, and ICMP traffic [18]. Another highlight of this study is that it generates assaults using genuine Kali Linux. Other studies just discuss assaults without actually performing them, thus this technique lends an added element of legitimacy to the study's conclusions. The accuracy of the results of this study is significantly superior to that of other studies, making it an important contribution to the current literature on inter-vlan routing. However, it should be noted that this

study only assesses SVI and RoAS setups; additional configurations may be assessed in future studies. There is no study comparing SVI with RoAS ,after completing a thorough analysis of available VLAN literature.

Study	Simulator	Traffic	Traffic method	Vlan	Metric	Result
[15]	OPNET	Elevated	FTP ,HTTP	SVI	Throughput	20000 bits/s
					Latency	0.00060 ms
[16]	OPNET	Excessive	FTP ,HTTP	SVI	Throughput	2200000 bits/s
					Latency	0.008 ms
[17]	OPNET Packet Tracer	Elevated	HTTP, FTP, DATABASE	SVI	Bandwidth	17408 bits/s
					Latency	0.001 ms
					jitter	0.0008 ms
[18]	Unknown	Elevated	ICMP	SVI	Latency	90 ms
[19]	GNS3	Below average	Ping	RoAS	Latency	20 ms
					Packet loss in 25 packet	9
[20]	Packet Tracer	Extremely low	Ping	RoAS	Latency	2 ms
					jitter	0 ms
[21]	Packet Tracer	Extremely low	Ping	RoAS	Latency	1 ms
					jitter	0 ms
[22]	GNS3	Reduced	Ping	SVI	Latency	0.3 ms
					jitter	0.2 ms
[23]	GNS3	Elevated	Ping on 50 PC	SVI	Latency	150 ms
					Packet loss	%1

Table 7 :Network Metrics Result in other studies

According to Table 7 , evaluation of prior research about the application of Switched Virtual Interface (SVI) and Router on a Stick (RoAS) technologies is provided in the table above. Latency, jitter, and throughput are the major measures used to assess the performance of these systems. Furthermore, the age of the simulator utilized in the research might have an impact on the outcomes. Older simulators may be restricted in functionality, have a slower processing speed, and may not adequately reflect real-world network circumstances. This might result in erroneous or unreliable findings, making it difficult to make educated judgments concerning network technology. This study employed a heavy traffic load and a current simulator, EVE-NG, to analyze the performance of Switched Virtual Interface (SVI) and Router on a Stick (RoAS). The utilization of a high traffic load allows for a more realistic assessment of these technologies' performance in real-world circumstances, while the contemporary simulator assures that the results are trustworthy and represent actual network conditions.

Metrics	RoAS (EVE-NG)	SVI (EVE-NG)	RoAS (GNS3)	SVI (GNS3)
Packet per second	Max = 3200 Min = 1750	Max = 31000 Min = 3000	Max = 2590 Min = 1410	Max = 10600 Min = 6400
Bandwidth utilization	11 Megabit/s	135 Megabit/s	10 Megabit/s	60 Megabit/s
Latency	Max = 60 ms Min = 4 ms	Max = 20 ms Min = 2 ms	Max = 40 ms Min = 3 ms	Max = 24 ms Min = 1 ms
Congestion	28500 B/s	2800000 B/s	26000 B/s	69000 B/s
Jitter	29 m/s	9 ms	21ms	12 ms
Packet loss in 120 packet	14	7	38	4
Throughput	4900000 bits/s	95000000 bits/s	4800000 bits/s	36000000 bits/s
Download speed(on PC)	590 kb/s	9700 kb/s	560 kb/s	3200 kb/s

Table 8 :SVI vs RoAS Network Metrics Result in this Study

According to Table 8, the result of this inquiry provides a greater number of network metrics and specific details related to the comparison of SVI and RoAS. Moreover, the accuracy of this investigation is 99%. Based on the findings, it was determined that Switched Virtual Interface (SVI) is far more suited for inter-VLAN routing than Router on a Stick (RoAS).

Metrics	Router on a stick	Switched virtual interface	References
Bandwidth utilization	Capable of handling a limited quantity of data flow across the network	Has the ability to handle a substantial amount of data flow through the network	[6] Ali, Mohsin & Jehangiri, Ali & Alramli, Omar & Ahmad, Zulfiqar & Ghoniem, Rania & Ala'anzy, Mohammed & Saleem, Romana. (2023). Performance and Scalability Analysis of SDN-Based Large-Scale Wi-Fi Networks. Applied Sciences. 13. 10.3390/app13074170.
Latency	latency is significantly greater	The latency is typically within normal range	[41] W. by: baeldung, "How to describe network performance?," Baeldung on Computer Science, https://www.baeldung.com/cs/bandwidth-packet-loss-latency-jitter .
Congestion	The risk of network congestion and bottlenecks is a concern	Possesses enhanced capabilities for managing network congestion	[30] Mansi, "Congestion control," Scaler Topics, https://www.scaler.com/topics/computer-network/tcp-congestion-control/ .
Jitter	Packets takes longer to be send	Packets takes faster to be send	[41] W. by: baeldung, "How to describe network performance?," Baeldung on Computer Science, https://www.baeldung.com/cs/bandwidth-packet-loss-latency-jitter .
Packet loss	The system is seeing an increase in the amount of lost packets.	The system experiences a lower number of lost packets.	[41] W. by: baeldung, "How to describe network performance?," Baeldung on Computer Science, https://www.baeldung.com/cs/bandwidth-packet-loss-latency-jitter .

Throughput	The system does not have the capacity to process additional tasks within a given timeframe.	Has the capacity to process a greater number of tasks within a specified timeframe.	[42] A. Saxena, "Throughput in computer networks," Scaler Topics, https://www.scaler.com/topics/throughput-in-computer-networks/ .
Download speed	The router's download speed is significantly slower	The switch provides a significantly faster download speed	[43] "What different internet speeds mean," Go to Dorset Council, https://www.dorsetcouncil.gov.uk/business-consumers-licences/digital-dorset/what-is-fibre-broadband/what-different-internet-speeds-mean .
Scalability	The system is not equipped to handle additional traffic	Capable of accommodating increased traffic volume	[33] N. Security, Scalability. [Online]. Available: https://www.networksecurity.org/members-area/glossary/s/scalability.html .
Reliability	The system operates at a typical or average level of reliability	Offers a higher degree of reliability	[34] A. L. DeCarlo, "What's the difference between network reliability and availability?: TechTarget," Networking, https://www.techtarget.com/searchnetworking/answer/Whats-the-difference-between-network-availability-and-reliability .
Cost	The cost of the router is substantial	The cost of the switch is comparatively lower	[12] Cisco router, Cisco Switch, new used Cisco prices comparison, https://www.router-switch.com/ .
Security	Offers a higher level of security	Provides a level of performance or quality that is nearly comparable to that of a router	[40] Implementing Switch Port Security 106 Chapter 6 CCNA 200-301 volume 2

Table 9 :SVI vs RoAS Explained

Network Metrics	Router on a stick	Switched virtual interface
Packet per second	Low	High
Bandwidth utilization	Low	High
Latency	High	Normal
Congestion	High	Low
Jitter	High	Low
Packet loss	High	Low
Throughput	Low	Very high
Download speed	Slower	Faster
Scalability	Satisfactory	Excellent
Reliability	Satisfactory	Excellent
Cost	Expensive	less expensive
Security	Marginally superior	Acceptable

Table 10 :SVI vs RoAS

Table 9 and 10 above depict the results of tests undertaken to evaluate the performance of a network in the context of inter-VLAN routing. While the trials were conducted in settings that closely resembled real-world scenarios, the results are only applicable to inter-VLAN routing and may not be predictive of network performance in other contexts. It is critical to recognize that network performance is affected by a number of factors, including network architecture, hardware and software settings, and the protocols used. As a result, any evaluation of network performance must take these variables and their possible influence into account.

6. Conclusion and Recommendations

6.1 Conclusion

In conclusion, our study has shown that adopting Switched Virtual Interface (SVI) over Router on a Stick (RoAS) offers unique advantages in many circumstances, notably in terms of performance and flexibility. It is crucial to remember, however, that in terms of security, a router is frequently the preferable option. We were able to evaluate and compare the performance of these two network design techniques using network simulation tools such as GNS3 and EVE-NG. According to our findings, EVE-NG provided a superior testing environment and more reliable results. Finally, the decision between SVI and RoAS is determined by the network's individual demands and requirements, as well as the level of security required. It is critical to properly assess.

6.2 Limitation and Recommendations

Limitations:

While our study has offered useful insights into the benefits and drawbacks of switched virtual interface (SVI) and router on a stick (RoAS) network topologies, several limitations should be addressed. Our study has one restriction in that it was carried out using network modeling software rather than real-world implementations. As a result, the findings may not accurately reflect how certain network architectures behave in real-world circumstances. Furthermore, our research solely looked at the security implications of employing a router vs. a switch for inter-VLAN routing. Other security issues that were not included in our research may exist.

Recommendations:

Based on our findings, we would advise network managers to thoroughly assess their network's individual demands and requirements in order to determine if SVI or RoAS is the preferable option. SVI may be a preferable alternative for networks that value performance and flexibility, whereas networks with strong security needs may benefit more from employing a router for inter-VLAN routing. A future study might be undertaken utilizing real-world implementations rather than network modeling tools to acquire more realistic results. Furthermore, future studies might build on our findings by looking at additional security criteria that may be crucial when deciding between SVI and RoAS network topologies.

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Appendix

Configuring Router on a Stick

```
Router(config)# interface Ethernet0/1
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# interface Ethernet0/1.11
```

```
Router(config-subif)# encapsulation dot1Q 11
```

```
Router(config-subif)# ip address 192.168.11.11 255.255.255.0
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

```
Router(config)# interface Ethernet0/1.12
```

```
Router(config-subif)# encapsulation dot1Q 12
```

```
Router(config-subif)# ip address 192.168.12.12 255.255.255.0
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

```
Router(config)# interface Ethernet0/1.13
```

```
Router(config-subif)# encapsulation dot1Q 13
```

```
Router(config-subif)# ip address 192.168.13.13 255.255.255.0
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

```
Router(config)# interface Ethernet0/1.14
```

```
Router(config-subif)# encapsulation dot1Q 14
```

```
Router(config-subif)# ip address 192.168.14.14 255.255.255.0
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

```
Router(config)# interface Ethernet0/1.15
```

```
Router(config-subif)# encapsulation dot1Q 15
```

```
Router(config-subif)# ip address 192.168.15.15 255.255.255.0
```

```
Router(config-subif)# no shutdown
```

```
Router(config-subif)# exit
```

Switched Virtual Interface

Creating Vlans

```
switch(config)# vlan 11
switch(config)# vlan 12
switch(config)# vlan 13
switch(config)# vlan 14
switch(config)# vlan 15
```

Giving ip to vlan interfaces

```
switch(config)# interface vlan 11
switch(config-if)# ip address 192.168.11.11 255.255.255.0
switch(config-if)# no shutdown
switch(config-if)# exit
```

```
switch(config)# interface vlan 12
switch(config-if)# ip address 192.168.12.12 255.255.255.0
switch(config-if)# no shutdown
switch(config-if)# exit
```

```
switch(config)# interface vlan 13
switch(config-if)# ip address 192.168.13.13 255.255.255.0
switch(config-if)# no shutdown
switch(config-if)# exit
```

```
switch(config)# interface vlan 14
switch(config-if)# ip address 192.168.14.14 255.255.255.0
switch(config-if)# no shutdown
switch(config-if)# exit
```

```
switch(config)# interface vlan 15
switch(config-if)# ip address 192.168.15.15 255.255.255.0
switch(config-if)# no shutdown
switch(config-if)# exit
switch(config)# ip routing
```

Creating Vlan in all switches

```
switch(config)# vlan 11
switch(config)# vlan 12
switch(config)# vlan 13
```

```
switch(config)# vlan 14
```

```
switch(config)# vlan 15
```

Trunking ports in Switch

All ports in switch (SW0 , SW1, SW2, SW3 , and Switch L 3)

```
Switch(config)# interface range Ethernet0/0-3
```

```
Switch(config-if-range)# switchport trunk encapsulation dot1q
```

```
Switch(config-if-range)# switchport mode trunk
```

```
Switch(config-if-range)# no shutdown
```

```
Switch(config-if-range)# exit
```

Accessing ports to Vlan

In SW7

```
Switch(config)# interface e0/0
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 11
```

In SW6

```
Switch(config)# interface e0/0
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 12
```

In SW5

```
Switch(config)# interface e0/0
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 13
```

In SW4

```
Switch(config)# interface e0/0
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 14
```

```
Switch(config)# interface e0/3
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 14
```

In SW1

```
Switch(config)# interface e0/0
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 15
```